

Shorting-Fee Dynamics under Heterogeneous Beliefs^{*}

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Abstract

How do short-selling frictions interact with heterogeneous beliefs to shape short exposure and asset-price dynamics? We document an empirical puzzle: borrowing fees rise gradually to a peak and then decline, yet stock prices begin falling well before the fee peak and continue to underperform afterward. To study these joint dynamics, we develop a dynamic general equilibrium model with heterogeneous beliefs where the cost of maintaining a short position increases with its size. Separating physical lending capacity from a convex carrying cost allows short exposure to adjust smoothly. We show that the carrying-cost coefficient primarily disciplines the quantity of short exposure rather than its equilibrium borrowing fee. Because lending capacity remains slack along the equilibrium path, the increasing marginal carrying cost provides the restoring force that generates mean reversion in short exposure. The model captures the qualitative features of this empirical trajectory, establishing a conceptual framework for the co-evolution of asset prices and borrowing costs.

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1 Introduction

How do short-selling frictions interact with investor beliefs to shape asset prices and the dynamics of short exposure? In frictionless markets, short selling provides a mechanism for investors who disagree with prevailing valuations to trade against overpricing. In actual securities markets, however, maintaining a short position requires borrowing shares and paying a cost that can vary substantially across securities and over time. When borrowing conditions become tight, fees¹ can rise sharply even as the underlying stock is already declining. This raises a basic theoretical question: *does the cost of carrying a short position merely distort prices, or does it also generate the restoring force that brings both the position and the price back toward a stable level?*

We motivate this question with a distinct empirical regularity in the dynamics of equity fee-spike episodes. While shorting fees remain low for most securities most of the time, with 62.4% of daily observations sitting below 1% per annum, they spike aggressively when shorting activities become unusually intense. As shown in [Figure 1](#), tracking the average chronological path reveals a puzzling timeline mismatch. Borrowing fees typically rise gradually ahead of these events, rather than spiking immediately, and then decline. Meanwhile, stock prices begin falling before these fees peak. Even after the fees drop from their high point, the affected stocks continue to underperform the broader market.

This timing pattern is hard to explain if borrowing costs depend mostly on current fundamentals or stock prices. Instead, it highlights the importance of the accumulated short exposure itself. Because short positions build up gradually, the cost to maintain them can rise as they grow larger. This accumulated short position acts as a distinct variable, decoupling the timing of borrowing costs from actual asset prices. Consequently, when fundamentals reverse, the short position doesn't always adjust right away. This delay allows borrowing

¹This paper uses *shorting fee*, *borrowing fee* and *lending fee* interchangeably for the per-share borrowing charge scaled by price, f_t/p_t^s . In the model, this corresponds to the borrowing rate. See [Appendix A](#) for terminology and the mapping between the model and data. With a slack lending constraint throughout the economy, the base rate is zero, meaning this fee is identical to the size premium as a rate on price, $\frac{1}{2}\gamma\theta_{t+1}^P/p_t^s$.

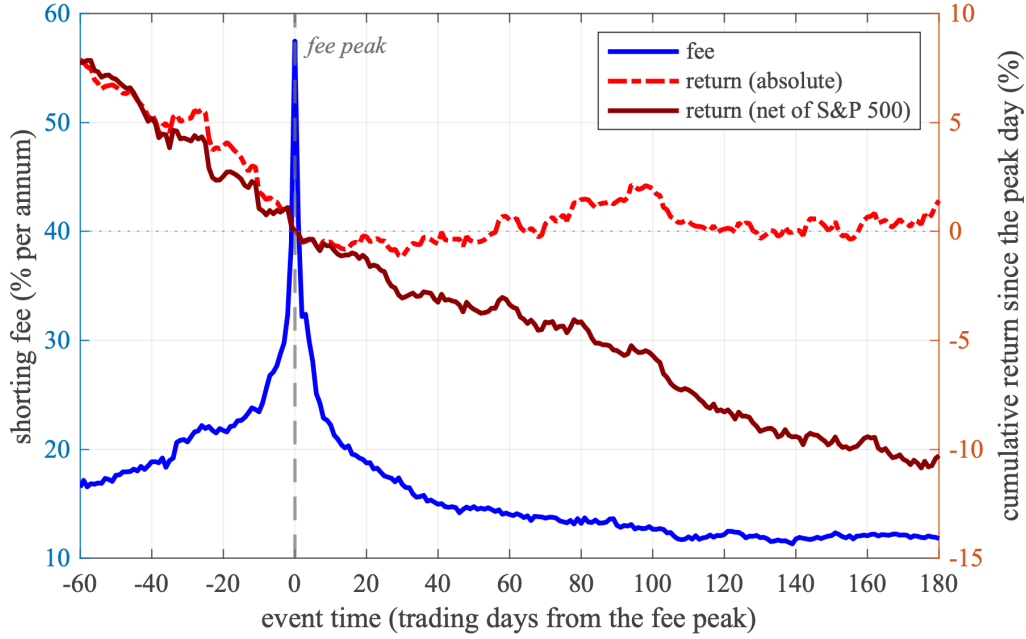


Figure 1: Dynamics of Shorting Fees and Stock Returns Around Fee-Spike Episodes.

The solid blue line shows the average indicative shorting fee (% per annum; left axis). The red dash-dotted line and dark-red solid line show average cumulative returns on the security and on the security in excess of the S&P 500, respectively (right axis). Returns are measured relative to the stock price on the fee-peak day, so positive values before Day 0 indicate that the stock price was above its event-day level. The horizontal axis reports event time in trading days over a 241-day window, from 60 days before to 180 after the shorting-fee peak. Day 0 denotes the fee peak and is marked by the vertical dashed line. An episode requires a fee peak above 10% APR; [Figure B.1](#) repeats the composite across alternative cutoffs, from no threshold to 50%. Source: `code/plot_fee_return.m`.

conditions to stay tight even after the asset price has started moving in the opposite direction.

To study this mechanism, we develop a dynamic general equilibrium model featuring heterogeneous beliefs, short selling, and securities lending. The economy consists of an Optimist and a Pessimist who disagree on the long-run mean of a persistent dividend process. In this setup, the Pessimist can short sell the asset, but expanding this position becomes progressively more expensive due to a convex, position-dependent carrying cost governed by a specific intensity parameter. We explicitly distinguish this friction from a physical lending-capacity constraint. Whereas the capacity constraint defines the outer boundary of feasible short exposure, the carrying cost dictates the marginal expense of increasing that exposure; in turn, the resulting equilibrium borrowing rate determines the actual market price paid to short the asset.

The model delivers three main results. First, the position-dependent carrying cost introduces an endogenous restoring force for short exposure. As the position expands, its rising marginal cost weakens the incentive for further expansion and induces a contraction when the mispricing opportunity narrows. Because physical lending capacity never binds along our simulated equilibrium paths, this restoring force operates entirely through the escalating cost of maintaining the position rather than a hard quantity constraint. Consequently, the same friction that limits arbitrage also drives mean reversion in short exposure. Ultimately, through its effect on equilibrium short positions, this restoring force contributes to the eventual stabilization of asset prices following the decline in raw returns observed empirically.

Second, the intensity of this carrying-cost primarily drives quantities rather than the borrowing rate. Higher carrying-cost intensity substantially reduce the equilibrium amount of short exposure while having only a small effect on the equilibrium borrowing rate, indicating that the friction mainly affects how much investors short rather than the fee rate paid to borrow.

Third, the model explains the delayed response observed in borrowing fees. Because short exposure adjusts gradually, it functions as a slow-moving state variable that permits

short positions to accumulate over time. When fundamentals reverse, the asset price reacts immediately, whereas the accumulated short position adjusts with a lag. Consequently, the model generates a distinct endogenous timeline: the turning point in the asset price precedes the inflection point in short exposure, which in turn precedes the peak borrowing fee. This sequence rationalizes the empirical finding that borrowing fees continue to rise even after the stock price has begun to decline.

The quantitative analysis also reveals a limitation of this mechanism. The smooth, position-dependent carrying cost captures the qualitative delayed response of borrowing fees to movements in the stock price, but it does not fully reproduce the magnitude of the observed fee spikes. This limitation suggests that a modification or additional sources of variation in borrowing conditions (such as discrete lending constraints, lender recalls, institutional participation, or search and matching frictions) may be important for understanding extreme borrowing-cost episodes.

The remainder of this paper is organized as follows. [Section 2](#) reviews the related literature. [Section 3](#) documents the data and stylized facts. [Section 4](#) develops the economic model. [Section 5](#) describes the quantitative implementation. [Section 6](#) characterizes the equilibrium and restoring force. [Section 7](#) studies the model's dynamic implications and counterfactuals. [Section 8](#) concludes.

2 Literature Review

Our paper relates to three strands of literature: heterogeneous beliefs and asset pricing, short-sale constraints and limits to arbitrage, and securities lending and borrowing costs. The first establishes how disagreement can generate persistent valuation differences; the second studies how restrictions on short selling affect price discovery; and the third explains how borrowing costs arise from conditions in securities-lending markets. We connect these strands in a dynamic general-equilibrium framework in which belief disagreement generates short exposure, the marginal cost of shorting increases with position size, and the resulting portfolio dynamics feed back into equilibrium asset prices.

2.1 Heterogeneous Beliefs and Asset Prices

Our analysis builds on the literature on heterogeneous beliefs and asset pricing under trading restrictions. [Miller \(1977\)](#) shows that when short selling is constrained, disagreement need not be fully reflected in market prices because pessimistic investors cannot freely trade against the valuations of more optimistic investors. [Harrison and Kreps \(1978\)](#) and [Morris \(1996\)](#) provide foundational analyses of how belief disagreement and trading restrictions can generate prices that differ from benchmark fundamental valuations. [Scheinkman and Xiong \(2003\)](#) further show that heterogeneous beliefs, resale opportunities, and short-sale constraints can interact to generate persistent and nonlinear asset-price dynamics. [Dumas, Kurshev, and Uppal \(2009\)](#) study the asset-pricing implications of heterogeneous beliefs when investor sentiment and disagreement vary over time.

A related strand of the literature incorporates heterogeneous beliefs into general equilibrium and allows the distribution of wealth across investors to evolve endogenously. [Geanakoplos \(2010\)](#) shows how differences in beliefs, together with leverage constraints, can amplify asset-price fluctuations over the macroeconomic cycle. [Cao \(2018\)](#) studies how permanent belief heterogeneity affects the long-run distribution of wealth when investors face binding

financial constraints. These models emphasize that belief disagreement cannot be separated from the endogenous evolution of investors' wealth and trading opportunities. Our framework embeds this general-equilibrium perspective. Dividend realizations redistribute wealth between the Optimist and the Pessimist; in turn, this shifting wealth distribution dictates equilibrium prices and short-selling incentives. Consequently, short exposure and the wealth distribution evolve jointly over time.

Our paper differs from this literature in the way the limit to arbitrage is modeled. Standard frameworks represent short-sale restrictions as an exogenous or fixed constraint on the quantity of an asset that a pessimistic investor can short. Such a constraint determines whether and to what extent disagreement can be expressed through short selling, but it does not make the marginal cost of arbitrage depend continuously on the size of the position. We instead allow the cost of maintaining a short position to increase with aggregate short exposure through a convex, position-dependent carrying cost. The carrying-cost intensity parameter therefore affects the marginal incentive to expand short exposure rather than simply imposing a fixed upper bound on the position.

This formulation makes the short position itself an endogenous component of the equilibrium dynamics. As disagreement changes, the Pessimist adjusts her short exposure, which in turn affects the cost of maintaining the position and feeds back into equilibrium prices. This perspective complements the existing heterogeneous-beliefs literature by shifting attention from a binary distinction between constrained and unconstrained short selling toward the dynamic adjustment of short exposure under a position-dependent cost. It also provides a natural link to the securities-lending literature, where borrowing costs arise from the institutional conditions governing the supply and allocation of lendable securities.

2.2 Short-Sale Constraints and Limits to Arbitrage

A related literature examines how short-sale constraints affect price discovery and the capability of pessimistic investors to trade against overvaluation. [Jones and Lamont \(2002\)](#)

demonstrate that shorting frictions are associated with inflated stock prices and depressed subsequent returns, confirming that these restrictions delay the incorporation of negative information into prices. [Cohen, Diether, and Malloy \(2007\)](#) isolate shifts in equity lending markets to show that variation in shorting constraints explains the predictive power of shorting activity for future stock returns. Collectively, these foundational studies establish short-sale constraints as a primary limit to arbitrage that significantly distorts the price discovery process.

A broader empirical literature examines how institutional short-selling restrictions affect market efficiency and asset prices. [Beber and Pagano \(2013\)](#) study global short-selling bans during the 2007–2009 financial crisis, finding that these restrictions severely harmed market liquidity and price efficiency. Similarly, [Boehmer, Jones, and Zhang \(2013\)](#) investigate the 2008 U.S. shorting ban and demonstrate that shackling short sellers degrades price discovery while deeply impairing market quality. [Saffi and Sigurdsson \(2011\)](#) complement these event studies by analyzing the global relation between share supply and equity dynamics, showing that constrained securities-lending conditions systematically limit short-selling activity and retard price efficiency. Taken together, this literature establishes that the ability to short sell, the availability of lendable shares, and the resulting trading activity are vital determinants of how rapidly information is incorporated into asset prices.

Our framework is related to this literature but focuses on a different margin of the short-selling constraint. The studies above primarily emphasize a binary environment: whether short selling is permitted, how much of an asset can be shorted, or how the hard availability of shares caps shorting activity. In contrast, our model allows short exposure to adjust endogenously and introduces a position-dependent carrying cost that increases continuously with the size of the aggregate short position. The resulting constraint is therefore smooth rather than a fixed quantity restriction; investors can expand short exposure, but doing so becomes progressively more expensive. This distinction allows us to study not only whether short selling is constrained, but also how the escalating marginal cost of maintaining a

position shapes the equilibrium path and intertemporal evolution of short exposure.

By framing the friction this way, our model unifies the limits-to-arbitrage literature with the institutional mechanics of the securities-lending market. When short exposure is small, the marginal cost of expanding the position remains low, whereas larger aggregate positions generate a steep marginal carrying cost. The borrowing rate associated with the equilibrium position therefore reflects the continuous interaction between investor demand for short exposure and the economic cost of supplying and maintaining that exposure. This setup provides a dynamic link between short-sale constraints, borrowing costs, and asset prices that is entirely distinct from models relying on an exogenous short-sale bound.

2.3 Securities-Lending and Borrowing Costs

The securities lending literature studies the institutional mechanisms through which investors obtain shares to establish short positions and the borrowing costs associated with these transactions. [Duffie, Gârleanu, and Pedersen \(2002\)](#) develop a search-based model of securities lending in which differences in the availability of lendable shares and the process of matching borrowers and lenders affect equilibrium borrowing fees. Their framework highlights that shorting costs are determined not only by investors' beliefs about the underlying asset, but also by conditions in the securities lending market. Subsequent work examines how shortages of lendable securities can generate rents and affect equilibrium prices ([Duffie, 2010](#); [Blocher, Reed, and Van Wesep, 2013](#)), as well as the implications of these lending constraints for arbitrage strategies ([Mitchell, Pedersen, and Pulvino, 2007](#)).

A more recent literature emphasizes the dynamic interaction between shorting costs and arbitrage positions. [Gârleanu, Panageas, and Zheng \(2026\)](#) develop a dynamic framework in which variation in shorting costs affects arbitrage positions and asset-price dynamics. We share their interest in how shorting costs and short positions evolve together, but differ in the source and role of the borrowing-cost friction. Whereas their framework emphasizes a rebate channel, in which fee revenue passed to lenders can, under a sufficiently flat fee schedule, cause

short-selling demand to bend backward, we introduce a convex, position-dependent carrying cost that increases with the size of the aggregate short position. The two frictions also differ in what they produce: theirs admits multiple equilibria at the same wealth distribution, so utilization jumps as the economy crosses the region of multiplicity, whereas the solved policy here is continuous in the state, so the short position adjusts smoothly and mean-reverts within a bounded ergodic set.²

This formulation makes the marginal cost of shorting endogenous to the quantity of short exposure and generates an endogenous restoring force for the position itself. Crucially, it allows us to distinguish between two conceptually different constraints on short selling: physical lending capacity, which determines the boundary of feasible short exposure, and the position-dependent carrying cost, which determines the marginal cost of increasing the position within that boundary. Along the equilibrium paths studied in the paper, the physical lending constraint remains slack, so the borrowing rate is determined by the position-dependent carrying cost rather than by a binding quantity constraint.

This dynamic adjustment also connects our analysis to the broader literature on slow-moving capital and gradual arbitrage adjustment. Because short-selling positions adjust gradually in our model, borrowing conditions need not move contemporaneously with asset prices. Persistent changes in fundamentals allow short positions to accumulate over time, while a subsequent reversal can cause the asset price to turn before the accumulated short position has fully unwound. The resulting delay between movements in the asset price, short exposure, and the borrowing rate provides a mechanism for the empirical timing pattern documented in this paper.

Taken together, these literatures provide the building blocks for our analysis but leave open how a position-dependent cost of shorting shapes the joint dynamics of short exposure, borrowing costs, and asset prices. We contribute to these literatures by developing a framework that addresses this gap, making the cost of maintaining a short position endogenous to

²Because this curvature makes the marginal borrow rate linear in the position, the rebate channel cannot produce that backward bend in our setting; [Section 4.2](#) makes the comparison precise.

aggregate exposure within a dynamic general-equilibrium model. This specification allows us to distinguish the role of physical lending capacity from that of the position-dependent carrying cost and to study how the latter affects both the level and adjustment of short exposure.

3 Data and Stylized Facts

3.1 Data

We use a daily panel of indicative shorting fees, prices, and the S&P 500 benchmark covering 3,834 US tickers from 6 June 2022 to 6 December 2023, 1,412,161 ticker-days in total. The sample period is constrained by data availability and public restrictions on historical daily indicative-fee data used in this study. The relatively short sample period limits our ability to assess whether the documented dynamic relation is stable across different market regimes. Our empirical analysis therefore focuses on characterizing the relation between borrowing fees and stock prices within the observed sample period.

[Table 1](#) reports the panel’s coverage and the full fee distribution. The fee variable is the indicative annualized borrowing rate quoted for the security, reported throughout as an annual percentage rate (APR) and accrued daily, so a stock quoted at 57.5% APR costs roughly 0.23% of the position each trading day. From this panel we build an event sample: each ticker contributes at most one episode, dated by the day its own fee peaks, and an episode is kept only if that peak exceeds 10% per annum³ and the panel supplies 60 trading days before it and 120 (to 180) days after. That leaves the 405 episodes used throughout.⁴ To ensure our findings are not driven by this specific threshold, we later relax this filter and analyze the full spectrum of fee intensities, spanning cutoffs from zero up to 50% APR, demonstrating that the normalized structural fee-ramp of these episodes remains scale-invariant ([Appendix B.3](#)).

We use episodes rather than a cross-section, for two reasons. The first is the distribution in [Table 1](#): the median ticker-day is quoted at 0.46% per annum and 62.4% sit below 1%,

³The threshold is deliberately stricter than the conventional line between general collateral and specials, which the securities-lending literature draws near 1% APR. [D’Avolio \(2002\)](#) finds 91% of borrowed stock trading as general collateral at an average of 17 basis points, against 9% special at an average of 4.30%; a 10% cutoff therefore sits at roughly twice the average special, well inside the tail rather than at its boundary. It also coincides with the highest tier used in practitioner reporting, which buckets borrow costs as general collateral, under 1%, 1 to 5%, 5 to 10%, and 10% and above ([S3 Partners, 2024](#)).

⁴The attrition is mostly the post-window requirement (read [Appendix B.1](#)).

Statistic	Value
<i>Shorting fee, percent per annum</i>	
Median	0.46
75th percentile	3.89
90th percentile	15.24
95th percentile	32.75
99th percentile	104.41
Maximum	1188.52
<i>Tail shares, percent of ticker-days</i>	
Fee below 1% APR	62.4
Fee above 10% APR	13.9
<i>Coverage</i>	
Ticker-days	1,412,161
Tickers	3,834
Trading days	379

Table 1: The shorting-fee distribution and panel coverage. Percentiles and tail shares of the indicative shorting fee across every ticker-day, together with the size and span of the panel they are drawn from, before any episode filter is applied. [Figure 2](#) plots the same distribution; the table gives the values behind it. Source: `code/export_fee_summary_table.do`.

while the 99th percentile is 104% and the maximum 1,189%. Ordinary borrowing is close to free, so the variation this paper is about lives entirely in a thin tail that a cross-section would average away. The second is that the moments the model is judged against are ratios formed *within* an episode, the fee at its peak against its own level at $t - 60$ and at $t + 120$. Neither has a cross-sectional counterpart, because both need a time axis anchored on an event.

3.2 Facts

The panel yields the following stylized facts.

Fact 1: The fee ramps into the peak rather than jumping, falls back after.

Averaged across the 405 episodes, [Figure 1](#) shows the fee stands at 16.51% per annum sixty days before the peak and reaches 57.46% on the peak day, a ratio of 3.481. It is already elevated well before the spike and rises steadily into it, so an episode is a build over weeks rather than a single event. By $t + 120$, the average fee has fallen to 12.18%, or 0.212 of its

peak.

Fact 2: Prices fall into the peak and then underperform. Sixty days before the peak the stock stood 7.7% above its event-day level, so borrowing is most expensive while the price is falling and the fee reaches its maximum well after the decline has begun rather than at its onset. The stock does not recover afterwards: measured against the S&P 500 it gives up a further 8% – 10% on average over the following 120 – 180 trading days, while the raw price stays roughly flat near zero (normalized at fee-peak). [Section 7](#) traces their movements in the model, showing that the short carrying-cost still rises into a falling price and the price gives back its support only slowly.

Fact 3: The size of the shorting fee spike scales continuously with the magnitude of the subsequent price decline. This asset pricing anomaly does not depend on our arbitrary 10% threshold. Rebuilding the event study across cutoffs from 0% to 50% APR reveals that the return from $t - 60$ to $t + 120$ falls monotonically from -8.40% to -19.57% net of the S&P 500, and from -0.65% to -12.91% raw, while peak fees climb from 23.8% to 154.6% APR ([Table B.2](#)).

Fact 4: Borrowing is essentially costless under normal market conditions. The median ticker-day carries an annualized fee of just 0.46%, and 62.4% of all observations sit below 1%. For the vast majority of securities, shorting costs are practically negligible. Consequently, any microfounded friction a model imposes on shorting must remain nearly inactive within this dominant, low-fee regime.

Fact 5: and occasionally it is enormous. The same distribution has a very long right tail. The ninetieth percentile is 15.2% per annum, the ninety-ninth is 104%, and the maximum

⁵An order of magnitude is a factor of ten, so a tail three of them above one percent reaches roughly a thousand times that level: the largest fee in the panel is 1,189 times one percent and 2,588 times the median ticker-day. A range that wide cannot be shown on a linear axis without compressing the bulk of the distribution into an invisible sliver at the origin, which is why this panel uses a logarithmic one. That axis also cannot show the 46 ticker-days quoted at or below 0.01% APR.

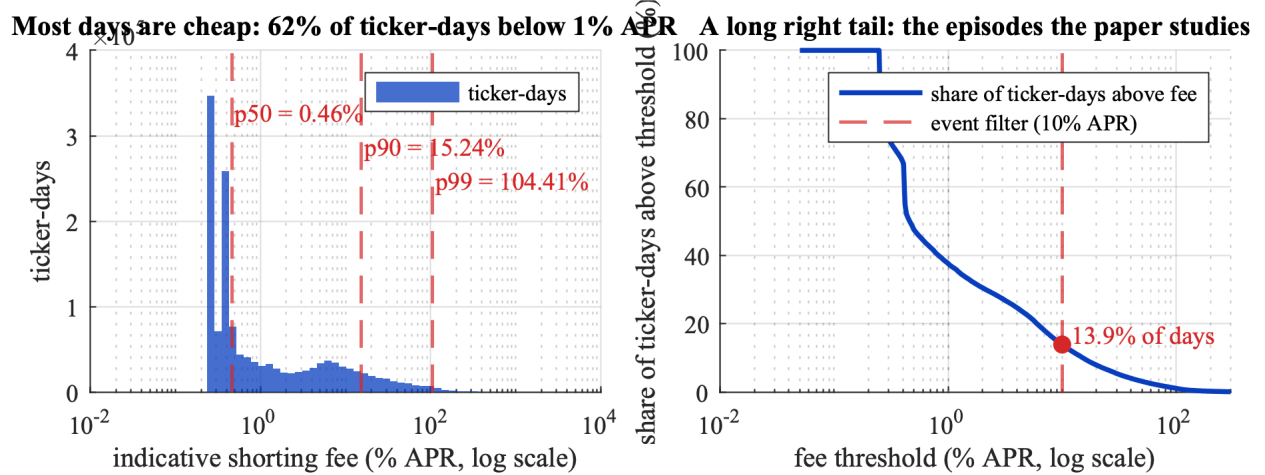


Figure 2: The Shorting Fee Across All 1.41 Million Ticker-Days. *Left:* the distribution of the indicative fee on a logarithmic axis. The mass is concentrated far below one percent per annum, with a right tail running three orders of magnitude higher.⁵ *Right:* the share of ticker-days on which the fee exceeds a given threshold. The dashed line marks the 10% APR cutoff: an episode requires a fee peak above it. Source: `code/plot_fee_distribution_facts.m`.

is 1,189%. Fees above 10% occur on 13.9% of ticker-days. [Facts 4](#) and [5](#) together are a strong joint restriction: the fee is essentially zero in normal times and can be large in stress episodes. In the model presented in [Section 4.6](#), this is a theorem rather than a calibration choice, as the fee (i.e., the base rate in our model) remains exactly zero unless lending capacity is fully exhausted.

Fact 6: Lendable supply is capped well below the float. Limited supply restricts short sellers to borrowing exclusively from shares enrolled in lending programs. [D’Avolio \(2002\)](#) reports that lendable supply averages roughly one-quarter to one-half of shares outstanding for US equities, a constraint driven almost entirely by the concentration of institutional ownership.⁶ This scarcity justifies the inventory cap $\ell_{t+1}^i \leq \kappa s_{t+1}^i$ embedded in the lending market setup ([Section 4.2](#)).

⁶Institutional ownership varies drastically across stocks, forming the primary source of borrowable float because institutions enroll holdings in lending programs as a matter of course. High institutional holding provides a deep, cheap borrowable float; low institutional ownership creates tight supply where the inventory cap binds. The model parameter κ captures this by representing a typical borrowable fraction within this empirical range rather than an immutable market constant.

Fact 7: Short interest can exceed the float. A share that is borrowed and sold is owned by its buyer, who may lend it again. Borrowing therefore compounds, and reported short interest can exceed 100% of shares outstanding without any accounting violation: in the January 2021 GameStop episode it reached roughly 140% of float. This is the fact behind the re-lending arithmetic that the model develops for the lending market in [Section 4.2](#), in which an inventory cap of κ supports a short of up to $\kappa/(1 - \kappa)$, and it is why the cap bounds the *quantity* of shorting the model can sustain rather than its price.

Scope. The fees are indicative quotes, not realized transaction costs from executed loans.⁷ What protects the exercise is that both benchmark fee moments are ratios of one quoted fee to another, so a bias common to the whole quoted path cancels and leaves both unchanged. [Appendix B.4](#) works that through, and isolates the bias.

⁷An indicative rate is the borrow cost a lending agent advertises for a security, quoted daily whether or not any loan is struck at it; a realized rate is what a borrower actually paid on an executed loan, observable only from the books of a lending intermediary, as in [D’Avolio \(2002\)](#). This panel carries the quote alone.

4 The Economic Model

This section develops a dynamic stochastic general equilibrium model to study the shorting-fee dynamic with heterogenous-belief agents. This is set out over the subsections that follow. [Section 4.1](#) sets up the environment: agents and preferences, the Lucas tree, endowments, the belief wedge, and the assignment of instruments to agents (i.e. stock borrowing vs lending). [Section 4.2](#) adds the lending market, together with the three frictions that act on it (inventory cap, the basic fee that prices it, and the carrying cost). [Section 4.3](#) writes each agent's budget constraint and [Section 4.4](#) defines a sequential competitive equilibrium (SCE). [Section 4.5](#) derives each agent's problem and its optimality conditions. [Section 4.6](#) and [Section 4.7](#) then studies the two types of shorting frictions at the model's equilibrium and their consequences: a basic fee driven by scarce capacity, and a carrying cost (premium) that scales with position size. [Section 4.8](#) recasts the whole as a recursive competitive equilibrium (RCE), and [Section 4.9](#) formulates the system of equations that [Section 5](#) works on.

4.1 Environment

The economy is based on a pure-exchange asset economy (Lucas tree model) in infinite discrete time, $t = 0, 1, 2, \dots$, with two agents (Pessimist vs Optimist), one traded claim (a unit supply) and one exogenous source of uncertainty (dividend). Because there is no bond alongside the tree, borrowing here means borrowing *shares* (stock) rather than goods, which is what puts a lending market at the centre of the paper. And because the dividend is the only thing that moves exogenously, the only other thing the economy has to keep track of is what the two agents' past trading has left them holding.

Therefore, this subsection introduces the standard heterogeneous-belief Lucas asset pricing tree model (unit supply, CRRA preferences, common discounting, a Markov dividend), where two agents hold different and permanent beliefs about that dividend. From that, the paper develops the model of shorting-fee dynamic by adding a lending market ([Section 4.2](#)).

Agents and preferences. Two infinitely-lived expected-utility maximizers, indexed $i \in \{P, O\}$ for *Pessimist* and *Optimist*, have CRRA preferences over consumption,

$$U^i = \mathbb{E}_0^i \left[\sum_{t=0}^{\infty} \beta^t u(c_t^i) \right], \quad u(c) = \frac{c^{1-\sigma}}{1-\sigma}, \quad \beta \in (0, 1), \sigma > 0, \quad (1)$$

where $\mathbb{E}^i$ is the expectation taken under agent i 's own subjective measure. Here σ is the coefficient of relative risk aversion, and under CRRA it is also the reciprocal of the elasticity of intertemporal substitution.

The Lucas tree. It provides a unit supply which pays an exogenous per-period dividend d_t taking values in a finite set $\{d_1, \dots, d_N\}$ with $d_1 < \dots < d_N$ and $d_1 > 0$, evolving on a first-order Markov chain. The tree trades at the ex-dividend price p_t^s , so a share carried into period t delivers d_t and can then be resold at p_t^s . Supply is fixed at one unit, and that is what gives the lending market something to be *scarce*: an agent who wants a negative position cannot manufacture what she sells, so the only inventory she can draw on is the other agent's.

Beliefs. Let $\Pi^i \in \mathbb{R}^{N \times N}$ denote agent i 's subjective transition matrix over the N dividend states, $[\Pi^i]_{jk} = \Pr^i(d_{t+1} = d_k \mid d_t = d_j)$. The disagreement is

$$\Pi^P \neq \Pi^O, \quad \text{both irreducible and aperiodic,} \quad (2)$$

where agents hold permanent, non-updating belief matrices ($\Pi^O \neq \Pi^P$) over dividend states, following [Harrison and Kreps \(1978\)](#) and [Cao \(2018\)](#). The Optimist shifts probability toward higher dividends, while the Pessimist holds the objective, data-generating truth, making the short seller the informed party.

Endowments. Each agent receives a deterministic labour endowment of the consumption good, constant over time, invariant to the dividend state, and equal across the two agents by

purpose:

$$Y_t^P = Y_t^O = Y \quad (3)$$

This setup of invariance helps to remove the endowment risk that would give either agent a hedging demand other than belief.

The belief wedge is the sole engine of trade. The two agents commonly share a utility function, a discount factor, a CRRA coefficient, an endowment. The disagreement in belief is the only asymmetry the environment leaves standing. Were it absent, both agents would value the tree under the same measure and neither would have reason to take the other side of it: there would be no short for a lending market to intermediate and no scarcity for a fee to price.

Who borrows and who lends. Disagreement dictates trading roles: the Optimist values the asset above market price and lends shares, while the Pessimist values it below and borrows to short sell. To align with this belief ordering, the model restricts each agent to their required instruments. The Optimist holds long and lending positions, whereas the Pessimist can do long and short positions.

Positions and timing. s_{t+1}^P and s_{t+1}^O are agents's gross long (the number of shares each agent want to buy), and θ_{t+1}^P is the Pessimist's gross short (shares borrowed and sold), while ℓ_{t+1}^O is the number of shares the Optimist has lent out, all chosen at t . Prices, dividends and consumption are dated by the period in which they are paid: p_t^s , d_t and c_t^i .

The primitives are now in the setup, but they are not yet a market. The Pessimist wants a negative position in a claim whose supply cannot expand, so the shares she sells have to come out of the Optimist's book. There needs a lending market for that, therefore, [Section 4.2](#) introduces it.

4.2 Lending Market: A Quantity, a Price, and a Slope

Shorting requires the Pessimist to physically borrow shares from the Optimist’s inventory rather than recording a negative entry. Three distinct objects enter the model at different stages, representing three market frictions that define this paper’s specific lending environment.

The inventory cap: a quantity. A lender cannot lend her entire position because shares in cash accounts, pledged assets, or inactive programs reduce the lendable pool. The model operationalizes this scarcity by restricting an agent to lending at most a fixed fraction $\kappa \in (0, 1)$ of her holdings, entering as a constraint with its own multiplier. This is a quantity fact as described in [Fact 6](#) of Section 3.

The fee: the price of that quantity. The basic fee p_t^θ is the market-clearing price for borrowed shares and enters agent budgets as a direct transfer, and it may be zero. Zero is not a degenerate case to be assumed away but the ordinary one, since a lending market with inventory to spare has nothing to charge for, and [Fact 4](#) of Section 3 records that borrowing is in fact free almost all of the time.

What *fee* means here, and what it can mean elsewhere. In this paper, the basic fee p_t^θ denotes exclusively the per-share charge for the loan. It differs from three alternative definitions in the literature and market: first, the spread between the short rate and the cash collateral rebate rate (which turns negative for hard-to-borrow stocks); second, non-transacted indicative quotes; and third, the borrower’s all-in carrying cost, which this model isolates in a separate premium term.

The carrying cost: the part of the rate that grows with the short position. While most stocks are general collateral, borrowable in quantity at a few basis points, scarce supply forces stocks “on special”, where rates rise progressively with loan size. The model captures this by adding a convex term, $\frac{1}{2}\gamma(\theta_{t+1}^P)^2$, to the borrower’s budget, meaning larger positions

face higher marginal rates. This reflects empirical episodes where accrued borrow costs grow large enough to rival or exceed the underlying price movements.

Total charges: basic fee and carrying cost. Combining both charges, shorting θ_{t+1}^P shares costs the Pessimist

$$B_t(\theta_{t+1}^P) = \underbrace{p_t^\theta \theta_{t+1}^P}_{\text{basic fee}} + \underbrace{\frac{1}{2}\gamma(\theta_{t+1}^P)^2}_{\text{carrying cost}} \quad (4)$$

in total. Dividing by the quantity yields the quoted per-share rate,

$$f_t(\theta_{t+1}^P) = \frac{B_t(\theta_{t+1}^P)}{\theta_{t+1}^P} = \underbrace{p_t^\theta}_{\text{base rate}} + \underbrace{\frac{1}{2}\gamma\theta_{t+1}^P}_{\text{size premium}}, \quad (5)$$

while differentiating yields the marginal cost,

$$B'_t(\theta_{t+1}^P) = p_t^\theta + \gamma\theta_{t+1}^P. \quad (6)$$

The model intentionally distinguishes these pairs: the basic fee/carrying cost represent total budget deductions, whereas the base rate/size premium represent per-share rates. Crucially, because empirical data measures rates, the model is calibrated to match the total quoted rate f_t rather than the base rate p_t^θ alone.⁸

⁸The curvature chosen here also rules out the destabilizing mechanism of [Gârleanu, Panageas, and Zheng \(2026\)](#). That mechanism requires the wedge between what the borrower pays and what the lender receives, $h(y) \equiv f(y)(1 - \tau y)$ at utilization y , to decline somewhere. Their rebate parameter τ maps directly to χ here, so the channel is present in this model and nothing is switched off by assumption; what differs is the curvature of the schedule the rebate acts upon. A carrying cost quadratic in position size makes the marginal borrow rate linear in the position, and for a schedule of that shape $h'(y) > 0$ at every utilization and every pass-through rate, so the feedback loop never closes. Bending the demand curve backward instead requires a fee that stays flat while already positive, letting the subsidy to lenders outrun the marginal cost borne by borrowers. The two papers divide over the curvature of the fee schedule rather than the scale of the friction. Their flat supply curve also reduces lending-market clearing to a quadratic whose roots define the multiple equilibria, whereas the model here admits no closed form and is solved globally across the state space.

Lender only receives a fraction of total lending costs. Reflecting market practice, a short seller pays the full borrow rate while the beneficial owner receives only a fraction of the total costs because intermediaries retain the rest to cover operational and capital charges. In the model, the borrowing Pessimist pays $p_t^\theta + \frac{1}{2}\gamma(\theta_{t+1}^P)^2$, but the lending Optimist receives the full basic fee p_t^θ and only a fraction of the size premium, $\chi\frac{1}{2}\gamma(\theta_{t+1}^P)^2$, with the remainder going to an external intermediary.

4.3 Budget Constraints

Having defined the lending market through these three frictions — quantity (inventory cap), price (basic fee), and price-slope (size premium) — this subsection formalizes the constraints and positions they govern to establish the agents' budget constraints.

The Pessimist. Enters period t holding s_t^P long and θ_t^P short. She chooses consumption and a new position $(s_{t+1}^P, \theta_{t+1}^P)$ subject to the budget constraint:

$$c_t^P + p_t^s(s_{t+1}^P - \theta_{t+1}^P) + p_t^\theta \theta_{t+1}^P + \frac{1}{2}\gamma(\theta_{t+1}^P)^2 = (p_t^s + d_t)(s_t^P - \theta_t^P) + Y^P, \quad (7)$$

and non-negative positions:

$$s_{t+1}^P \geq 0, \quad \theta_{t+1}^P \geq 0, \quad (8)$$

where she may choose $\theta_{t+1}^P > s_{t+1}^P$ to be net short.

The Optimist. Enters holding s_t^O long, she chooses consumption, long and lent shares $(s_{t+1}^O, \ell_{t+1}^O)$ subject to a budget constraint showing she receives the base rate and a fraction χ of the carrying cost:

$$c_t^O + p_t^s s_{t+1}^O - p_t^\theta \ell_{t+1}^O - \chi\frac{1}{2}\gamma(\theta_{t+1}^P)^2 = (p_t^s + d_t) s_t^O + Y^O, \quad (9)$$

with non-negative positions:

$$s_{t+1}^O \geq 0, \quad \ell_{t+1}^O \geq 0, \quad (10)$$

and facing the inventory lending cap constraint:

$$\ell_{t+1}^O \leq \kappa s_{t+1}^O, \quad \kappa \in (0, 1). \quad (11)$$

4.4 Sequential Competitive Equilibrium

The established framework and environment enable an equilibrium definition as a stochastic process indexed by histories.

Definition 1 (Sequential competitive equilibrium). Let $d^t = (d_0, d_1, \dots, d_t)$ denote the history of dividend realizations up to period t . Given preferences and beliefs $(\beta, \sigma, \Pi^P, \Pi^O)$, endowments (Y^P, Y^O) , friction parameters (κ, γ) , the pass-through χ and initial stock asset holdings $(s_0^P, \theta_0^P, s_0^O)$ satisfying $s_0^P - \theta_0^P + s_0^O = 1$, a *sequential competitive equilibrium* is the mapping from histories d_t to:

1. Price functions: stock prices $p_t^s(d^t) > 0$ and lending fees $p_t^\theta(d^t) \geq 0$, and
2. Allocation functions: Consumption and portfolio choices $\{c_t^P(d^t), s_{t+1}^P(d^t), \theta_{t+1}^P(d^t)\}_{t \geq 0}$ for the Pessimist (P) and $\{c_t^O(d^t), s_{t+1}^O(d^t), \ell_{t+1}^O(d^t)\}_{t \geq 0}$ for the Optimist (O),

such that for every history d^t and at all period $t \geq 0$:

- (i) *Agents optimize.* Taking the price functions $p_t^s(d^t)$ and $p_t^\theta(d^t)$ as parametrically given, each agent $i \in \{P, O\}$ chooses their allocation to maximize expected lifetime utility as defined in (1):

$$\max \quad U^i = \mathbb{E}^{\Pi^i} \left[\sum_{t=0}^{\infty} \beta^t u(c_t^i(d^t)) \mid d^0 \right] \quad (12)$$

subject to:

- Pessimist's budget constraint (7), and non-negative positions (8).
- Optimist's budget constraint (9), non-negative positions (10), and lending cap (11).

(ii) *The share market clears.* Total long positions across both agents equal the fixed unit supply of the stock asset plus the total shorted shares by the Pessimist:

$$s_{t+1}^P(d^t) + s_{t+1}^O(d^t) = 1 + \theta_{t+1}^P(d^t).$$

(iii) *The lending market clears.* Shares borrowed equal shares lent,

$$\ell_{t+1}^O(d^t) = \theta_{t+1}^P(d^t).$$

Lemma 1 (The feasibility wall). Let prices and allocations satisfy clauses (i) to (iii) of

[Definition 1](#). Then at every history d^t , allocations are bounded by:

$$\theta_{t+1}^P \leq \bar{\theta} \equiv \frac{\kappa}{1 - \kappa}, \quad s_{t+1}^O \leq 1 + \bar{\theta}, \quad s_{t+1}^P \leq 1 + \bar{\theta}. \quad (13)$$

Every equilibrium position therefore lies in a bounded interval, naturally ruling out runaway short positions without requiring explicit no-Ponzi or transversality conditions.

Proof. The bound follows from three of the model's own conditions and the sign restriction on the Pessimist's long leg, in four steps.

Step 1: the short is what the Optimist lends. Loan clearing, clause (iii) of [Definition 1](#), is $\ell_{t+1}^O = \theta_{t+1}^P$. So a bound on her lending is a bound on the Pessimist's short.

Step 2: her lending is capped by her own book. The cap (11) is $\ell_{t+1}^O \leq \kappa s_{t+1}^O$. With Step 1,

$$\theta_{t+1}^P = \ell_{t+1}^O \leq \kappa s_{t+1}^O.$$

Step 3: her book is bounded by the float plus the short. Share clearing, clause (ii), is $s_{t+1}^P + s_{t+1}^O = 1 + \theta_{t+1}^P$, so $s_{t+1}^O = 1 + \theta_{t+1}^P - s_{t+1}^P$. The Pessimist's long leg is non-negative by

(8), and dropping it can only raise the right side:

$$s_{t+1}^O = 1 + \theta_{t+1}^P - s_{t+1}^P \leq 1 + \theta_{t+1}^P.$$

Step 4: solve the resulting inequality. Chaining Steps 2 and 3,

$$\theta_{t+1}^P \leq \kappa s_{t+1}^O \leq \kappa(1 + \theta_{t+1}^P),$$

and the short appears on both sides. Collecting it, $\theta_{t+1}^P(1 - \kappa) \leq \kappa$, and since $\kappa \in (0, 1)$ the factor $1 - \kappa$ is strictly positive, so dividing by it preserves the direction and delivers the first bound in (13). The remaining two follow from clause (ii): $s_{t+1}^O = 1 + \theta_{t+1}^P - s_{t+1}^P \leq 1 + \bar{\theta}$ because $s_{t+1}^P \geq 0$, and symmetrically $s_{t+1}^P = 1 + \theta_{t+1}^P - s_{t+1}^O \leq 1 + \bar{\theta}$ because $s_{t+1}^O \geq 0$ by (10).

Each of the three positions is therefore confined to a closed bounded interval, below by the non-negativity restrictions (8) and (10) and above by (13). $\square$

Remark 1 (The wall as a re-lending series). The feasibility wall emerges as the limit of a geometric series generated by compounding rounds of re-lending. Starting from a float of one share ($\theta_0 = 0$), each share lent is sold to a new buyer who can lend a fraction κ of it in turn. The short interest backing round $n + 1$ satisfies:

$$\theta_{n+1} = \kappa(1 + \theta_n). \tag{14}$$

Iterating from $\theta_0 = 0$ yields the closed-form sum $\theta_n = \bar{\theta}(1 - \kappa^n)$. Since $\kappa \in (0, 1)$, the sequence strictly increases to the fixed point of the system without ever reaching it in finite rounds:

$$\sum_{j=1}^{\infty} \kappa^j = \frac{\kappa}{1 - \kappa} = \bar{\theta}.$$

Two direct implications follow from $\kappa \in (0, 1)$:

1. The wall strictly exceeds the initial cap ($\bar{\theta} - \kappa = \frac{\kappa^2}{1 - \kappa} > 0$).

2. The wall exceeds 100% of the float ($\bar{\theta} > 1$) if and only if the cap passes one-half ($\kappa > \frac{1}{2}$).

This geometric compounding explains how short interest can exceed outstanding shares (Fact 7 of Section 3). Crucially, if $\kappa \rightarrow 1$, the series diverges; thus, the physical tightness of the inventory cap, rather than a terminal condition, is what prevents an infinite, runaway short position.

Remark 2 (Walras' law and deadweight loss). Goods market clearing is not independently imposed; it follows from clauses (i) to (iii) by Walras' law.

Proof. Summing the agents' budget constraints (7) and (9), the equity trading terms cancel via stock market clearing (ii). The base lending fee payments cancel via lending market clearing (iii) since $\theta_{t+1}^P = \ell_{t+1}^O$. Aggregating the remaining terms yields the economy's resource constraint:

$$c_t^P + c_t^O = d_t + Y^P + Y^O - (1 - \chi) \frac{1}{2} \gamma (\theta_{t+1}^P)^2. \quad (15)$$

The base rate p_t^θ and the shared friction fraction χ drop out of (15) because they represent pure internal transfers. The unrecovered portion of the carrying cost, $(1 - \chi) \frac{1}{2} \gamma (\theta_{t+1}^P)^2$, is retained by external intermediaries, functioning as an aggregate deadweight loss that destroys output whenever the short channel is active. $\square$

4.5 Agent Optimality Conditions

Once the definition of equilibrium is established ([Definition 1](#)), each agent's optimality condition can be derived by Lagrangians and F.O.C as follows:

The Pessimist. Her Lagrangian:

$$\begin{aligned} \mathcal{L}^P = \mathbb{E}_0^P \sum_{t \geq 0} \beta^t \Big\{ & u(c_t^P) + \lambda_t^P \left[(p_t^s + d_t)(s_t^P - \theta_t^P) + Y^P - c_t^P - p_t^s(s_{t+1}^P - \theta_{t+1}^P) \right. \\ & \left. - p_t^\theta \theta_{t+1}^P - \frac{1}{2} \gamma (\theta_{t+1}^P)^2 \right] \\ & \left. + \mu_t^{P,s} s_{t+1}^P + \mu_t^{P,\theta} \theta_{t+1}^P \right\}, \end{aligned} \quad (16)$$

and her F.O.C. gives:

$$[c_t^P]: 0 = u'(c_t^P) - \lambda_t^P \quad (17)$$

$$[s_{t+1}^P]: 0 = -\lambda_t^P p_t^s + \beta \mathbb{E}_t^P [\lambda_{t+1}^P (p_{t+1}^s + d_{t+1})] + \mu_t^{P,s} \quad (18)$$

$$[\theta_{t+1}^P]: 0 = \lambda_t^P p_t^\theta - \lambda_t^P p_t^\theta - \lambda_t^P \gamma \theta_{t+1}^P - \beta \mathbb{E}_t^P [\lambda_{t+1}^P (p_{t+1}^s + d_{t+1})] + \mu_t^{P,\theta} \quad (19)$$

Since $\lambda_t^P = u'(c_t^P) = (c_t^P)^{-\sigma} > 0$ by (17) and (1), (18) and (19) can be divided through by λ_t^P and every remaining λ replaced in the same way. The pair $\lambda_{t+1}^P / \lambda_t^P$ collapses to a rate of consumption growth, so write

$$\mathcal{M}_{t+1}^P \equiv \beta \left(\frac{c_{t+1}^P}{c_t^P} \right)^{-\sigma}, \quad \tilde{\mu}_t^{P,x} \equiv \frac{\mu_t^{P,x}}{\lambda_t^P} = \mu_t^{P,x} (c_t^P)^\sigma, \quad x \in \{s, \theta\}, \quad (20)$$

for the stochastic discount factor and for the multiplier on agent P 's constraint $\{s, \theta\}$ measured in units of period t consumption. The normalized system is:

$$[s_{t+1}^P]: 0 = -p_t^s + \mathbb{E}_t^P[\mathcal{M}_{t+1}^P(p_{t+1}^s + d_{t+1})] + \tilde{\mu}_t^{P,s} \quad (21)$$

$$[\theta_{t+1}^P]: 0 = p_t^s - p_t^\theta - \gamma \theta_{t+1}^P - \mathbb{E}_t^P[\mathcal{M}_{t+1}^P(p_{t+1}^s + d_{t+1})] + \tilde{\mu}_t^{P,\theta} \quad (22)$$

The Optimist. Her Lagrangian:

$$\begin{aligned} \mathcal{L}^O = \mathbb{E}_0^O \sum_{t \geq 0} \beta^t \Big\{ & u(c_t^O) + \lambda_t^O \left[(p_t^s + d_t) s_t^O + Y^O - c_t^O - p_t^s s_{t+1}^O \right. \\ & \left. + p_t^\theta \ell_{t+1}^O + \chi \frac{1}{2} \gamma (\theta_{t+1}^P)^2 \right] \\ & \left. + \mu_t^{O,s} s_{t+1}^O + \mu_t^{O,\ell} \ell_{t+1}^O + \mu_t^{O,\text{cap}} (\kappa s_{t+1}^O - \ell_{t+1}^O) \right\}, \end{aligned} \quad (23)$$

and her F.O.C. gives:

$$[c_t^O]: 0 = u'(c_t^O) - \lambda_t^O \quad (24)$$

$$[s_{t+1}^O]: 0 = -\lambda_t^O p_t^s + \beta \mathbb{E}_t^O[\lambda_{t+1}^O (p_{t+1}^s + d_{t+1})] + \mu_t^{O,s} + \kappa \mu_t^{O,\text{cap}} \quad (25)$$

$$[\ell_{t+1}^O]: 0 = \lambda_t^O p_t^\theta + \mu_t^{O,\ell} - \mu_t^{O,\text{cap}} \quad (26)$$

Since $\lambda_t^O = u'(c_t^O) = (c_t^O)^{-\sigma} > 0$ by (24) and (1), (25) and (26) divide through by λ_t^O in the same way, with $\mathcal{M}_{t+1}^O$ and $\tilde{\mu}_t^{O,s}$, $\tilde{\mu}_t^{O,\ell}$, $\tilde{\mu}_t^{O,\text{cap}}$ as in (20). The normalized system is:

$$[s_{t+1}^O]: 0 = -p_t^s + \mathbb{E}_t^O[\mathcal{M}_{t+1}^O(p_{t+1}^s + d_{t+1})] + \tilde{\mu}_t^{O,s} + \kappa \tilde{\mu}_t^{O,\text{cap}} \quad (27)$$

$$[\ell_{t+1}^O]: 0 = p_t^\theta + \tilde{\mu}_t^{O,\ell} - \tilde{\mu}_t^{O,\text{cap}} \quad (28)$$

Summary. The equilibrium system of all optimality conditions derived from [Definition 1](#) (sequential competitive equilibrium) presents as following:

The Equilibrium System

(the conditions implied by [Definition 1](#))

Euler conditions. With $\mathcal{M}_{t+1}^i = \beta(c_{t+1}^i/c_t^i)^{-\sigma}$ for $i \in \{P, O\}$ as in (20),

$$[s_{t+1}^P] \quad 0 = -p_t^s + \mathbb{E}_t^P[\mathcal{M}_{t+1}^P(p_{t+1}^s + d_{t+1})] + \tilde{\mu}_t^{P,s} \quad (\text{S1})$$

$$[\theta_{t+1}^P] \quad 0 = p_t^s - p_t^\theta - \gamma \theta_{t+1}^P - \mathbb{E}_t^P[\mathcal{M}_{t+1}^P(p_{t+1}^s + d_{t+1})] + \tilde{\mu}_t^{P,\theta} \quad (\text{S2})$$

$$[s_{t+1}^O] \quad 0 = -p_t^s + \mathbb{E}_t^O[\mathcal{M}_{t+1}^O(p_{t+1}^s + d_{t+1})] + \tilde{\mu}_t^{O,s} + \kappa \tilde{\mu}_t^{O,\text{cap}} \quad (\text{S3})$$

$$[\ell_{t+1}^O] \quad 0 = p_t^\theta + \tilde{\mu}_t^{O,\ell} - \tilde{\mu}_t^{O,\text{cap}} \quad (\text{S4})$$

Budget constraints.

$$c_t^P + p_t^s(s_{t+1}^P - \theta_{t+1}^P) + p_t^\theta \theta_{t+1}^P + \frac{1}{2}\gamma(\theta_{t+1}^P)^2 = (p_t^s + d_t)(s_t^P - \theta_t^P) + Y^P \quad (\text{S5})$$

$$c_t^O + p_t^s s_{t+1}^O - p_t^\theta \ell_{t+1}^O - \chi \frac{1}{2}\gamma(\theta_{t+1}^P)^2 = (p_t^s + d_t) s_t^O + Y^O \quad (\text{S6})$$

Complementary slackness.

$$\tilde{\mu}_t^{P,s} \geq 0, \quad s_{t+1}^P \geq 0, \quad \tilde{\mu}_t^{P,s} s_{t+1}^P = 0 \quad (\text{S7})$$

$$\tilde{\mu}_t^{P,\theta} \geq 0, \quad \theta_{t+1}^P \geq 0, \quad \tilde{\mu}_t^{P,\theta} \theta_{t+1}^P = 0 \quad (\text{S8})$$

$$\tilde{\mu}_t^{O,s} \geq 0, \quad s_{t+1}^O \geq 0, \quad \tilde{\mu}_t^{O,s} s_{t+1}^O = 0 \quad (\text{S9})$$

$$\tilde{\mu}_t^{O,\ell} \geq 0, \quad \ell_{t+1}^O \geq 0, \quad \tilde{\mu}_t^{O,\ell} \ell_{t+1}^O = 0 \quad (\text{S10})$$

$$\tilde{\mu}_t^{O,\text{cap}} \geq 0, \quad \kappa s_{t+1}^O - \ell_{t+1}^O \geq 0, \quad \tilde{\mu}_t^{O,\text{cap}}(\kappa s_{t+1}^O - \ell_{t+1}^O) = 0 \quad (\text{S11})$$

Market clearing.

$$\text{shares:} \quad s_{t+1}^P + s_{t+1}^O = 1 + \theta_{t+1}^P \quad (\text{S12})$$

$$\text{loans:} \quad \ell_{t+1}^O = \theta_{t+1}^P \quad (\text{S13})$$

That is 13 conditions in 13 unknowns: 2 prices (p_t^s, p_t^θ) , 2 consumptions (c_t^P, c_t^O) , 4 positions $(s_{t+1}^P, \theta_{t+1}^P, s_{t+1}^O, \ell_{t+1}^O)$, and 5 multipliers $(\tilde{\mu}_t^{P,s}, \tilde{\mu}_t^{P,\theta}, \tilde{\mu}_t^{O,s}, \tilde{\mu}_t^{O,\ell}, \tilde{\mu}_t^{O,\text{cap}})$.

4.6 Basic Fee Is the Price of Scarce Lending Capacity

The [equilibrium system](#) of [Section 4.5](#) implies the following result:

Proposition 1 (Basic fee is the price of the cap). Let the Optimist's lending choice satisfy the inventory cap $\ell_{t+1}^O \leq \kappa s_{t+1}^O$ (11) at which she lends a strictly positive quantity $\ell_{t+1}^O > 0$. Then, at the equilibrium, a strictly positive fee can stand only against a cap held with equality:

$$p_t^\theta > 0 \iff \text{the inventory cap binds: } \ell_{t+1}^O = \kappa s_{t+1}^O. \quad (29)$$

Equivalently, capacity to spare, $\ell_{t+1}^O < \kappa s_{t+1}^O$, carries $\tilde{\mu}_t^{O,\text{cap}} = 0$ and so a zero fee, $p_t^\theta = 0$.

Proof. Her lending condition (S4) rearranges to:

$$\tilde{\mu}_t^{O,\text{cap}} = p_t^\theta + \tilde{\mu}_t^{O,\ell}. \quad (30)$$

Suppose $p_t^\theta > 0$. Since $\tilde{\mu}_t^{O,\ell} \geq 0$, (30) gives $\tilde{\mu}_t^{O,\text{cap}} \geq p_t^\theta > 0$, and the cap's complementary slackness $\tilde{\mu}_t^{O,\text{cap}}(\kappa s_{t+1}^O - \ell_{t+1}^O) = 0$ from (S11) then forces $\kappa s_{t+1}^O - \ell_{t+1}^O = 0$. Conversely, if the cap binds $\kappa s_{t+1}^O - \ell_{t+1}^O = 0$, then (S11) sets $\tilde{\mu}_t^{O,\text{cap}} > 0$. And given $\ell_{t+1}^O > 0$ set $\tilde{\mu}_t^{O,\ell} = 0$, so (30) collapses to: $p_t^\theta = \tilde{\mu}_t^{O,\text{cap}} > 0$. $\square$

Remark 3 (Basic fee as a complementarity). [Proposition 1](#) can be stated as complementarity:

$$p_t^\theta \geq 0, \quad \kappa s_{t+1}^O - \ell_{t+1}^O \geq 0, \quad p_t^\theta (\kappa s_{t+1}^O - \ell_{t+1}^O) = 0, \quad (31)$$

which is exact here as a property of equilibrium.

The lending yield in the share price. The identity (30) behind the proof also determines what a share costs the agent who supplies the loan. Represented as follows:

Corollary 1 (The share price carries a lending yield). At any history at which the Optimist

lends a strictly positive quantity, $\ell_{t+1}^O > 0$, then her long-leg condition reads:

$$p_t^s - \kappa p_t^\theta = \mathbb{E}_t^O[\mathcal{M}_{t+1}^O(p_{t+1}^s + d_{t+1})]. \quad (32)$$

Proof. Since $\ell_{t+1}^O > 0$ (her lending floor is slack) and $\tilde{\mu}_t^{O,\ell} \ell_{t+1}^O = 0$ from (S10) sets $\tilde{\mu}_t^{O,\ell} = 0$, so (30) reads $\tilde{\mu}_t^{O,\text{cap}} = p_t^\theta$. The inventory cap $\ell_{t+1}^O \leq \kappa s_{t+1}^O$ (11) also implies $s_{t+1}^O > 0$, then $\tilde{\mu}_t^{O,s} s_{t+1}^O = 0$ from (S9) sets $\tilde{\mu}_t^{O,s} = 0$. Substituting both into her long-leg condition (S3) and rearranging delivers (32). $\square$

Intuitively, the lender does not pay for a share at full cost p_t^s . She pays $p_t^s - \kappa p_t^\theta$, the price net of the basic fee income the share will earn, because each share she holds supports κ lendable units and the inventory cap (11) ties that capacity to her own book. So the basic fee income is netted off the purchase price. Read the other way round, (32) splits the quoted price into what she thinks the dividends are worth and what the lending capacity adds:

$$p_t^s = \underbrace{\mathbb{E}_t^O[\mathcal{M}_{t+1}^O(p_{t+1}^s + d_{t+1})]}_{\text{her valuation of the dividends}} + \underbrace{\kappa p_t^\theta}_{\text{the value of the lending capacity}}. \quad (33)$$

This is the model's version of the lending yield priced into hard-to-borrow stocks: a security that is expensive to borrow is worth more to hold than the claim to its dividends alone, since holding it is the only way to supply the loan, and the channel by which a price embeds the value of the loans the security will support is the subject of [Duffie, Gârleanu, and Pedersen \(2002\)](#).⁹

⁹The channel is the same; the market structure is not. There the basic fee is a bargaining outcome between a borrower and a lender who have found each other through search, the scarcity that gives it a positive value is the slow arrival of lendable supply, and the price ends up above the valuation of even the most optimistic investor, which delivers a predictable decline as more shares become lendable. Here the basic fee is a competitive price clearing the loan market, the scarcity is the inventory cap (11), and [Proposition 1](#) makes the basic fee that cap's shadow price. [Corollary 1](#) is that same decomposition read off a budget constraint: read recursively, it makes the price the Optimist's own valuation of the dividends plus the present value of the fees her share will earn. What differs is a friction of the second lending cost in our model. Their paper's short pays the basic fee as a lending agreement and viewed as "specialness", whereas here the borrower pays the lending capacity fee (the basic) and the convex carrying cost $\frac{1}{2}\gamma(\theta_{t+1}^P)^2$ (the premium) which grows with the position.

4.7 Two-Wedge Identity: Level Versus Slope

This subsection determines the total friction on the short (“total wedge”), and draws the consequence: Pessimist not holding long and short simultaneously.

In the [equilibrium system](#) of [Section 4.5](#), the Pessimist’s FOCs supply conditions on her long- and short-leg, (S1) and (S2), restated here with $V_t^P \equiv \mathbb{E}_t^P[\mathcal{M}_{t+1}^P(p_{t+1}^s + d_{t+1})]$ denoting her own valuation of the dividends¹⁰:

$$0 = -p_t^s + V_t^P + \tilde{\mu}_t^{P,s} \quad (\text{S1})$$

$$0 = p_t^s - p_t^\theta - \gamma \theta_{t+1}^P - V_t^P + \tilde{\mu}_t^{P,\theta} \quad (\text{S2})$$

that leads to a proposition below:

Proposition 2 (The two-wedge identity). At every history, the total wedge on the Pessimist’s short, measured by the sum of her non-negativity multipliers, is the marginal cost of shorting:

$$\tilde{\mu}_t^{P,s} + \tilde{\mu}_t^{P,\theta} = \gamma \theta_{t+1}^P + p_t^\theta. \quad (34)$$

Dividing through by the share price states the same wedge per unit of value shorted:

$$\underbrace{\frac{\tilde{\mu}_t^{P,s} + \tilde{\mu}_t^{P,\theta}}{p_t^s}}_{\text{total wedge on the short}} = \underbrace{\frac{\gamma \theta_{t+1}^P}{p_t^s}}_{\text{cost wedge}} + \underbrace{\frac{p_t^\theta}{p_t^s}}_{\text{basic fee wedge}}. \quad (35)$$

The two wedges are the two terms on the right: a carrying cost that rises with the position at rate γ (a *slope*), and a fee that is flat in it (a *level*).

Proof. Simply adding (S1) and (S2), then rearranged yields (34). For the latter, since $\mathcal{M}_{t+1}^P > 0$, $d_{t+1} > 0$, $p_{t+1}^s \geq 0$, $\tilde{\mu}_t^{P,s} \geq 0$, then (S1) bounds p_t^s below by $\mathbb{E}_t^P[\mathcal{M}_{t+1}^P d_{t+1}] > 0$,

¹⁰Measured in units of period t consumption, originally normalized by dividing $\beta \mathbb{E}_t^P[\lambda_{t+1}^P(p_{t+1}^s + d_{t+1})]$ through by λ_t^P which turns the discounted marginal utility λ_{t+1}^P into the stochastic discount factor $\mathcal{M}_{t+1}^P$, and leaves V_t^P in goods rather than utils, comparable to p_t^s and p_t^θ directly.

so that dividing (34) through by p_t^s gives (35). □

What the wedges measure. Rearranging (S1) as $\tilde{\mu}_t^{P,s} = p_t^s - V_t^P$ which is the market price less what the Pessimist herself thinks a share is worth, and defined as what one more share held long would cost her. While $\tilde{\mu}_t^{P,\theta} = (V_t^P + p_t^\theta + \gamma \theta_{t+1}^P) - p_t^s$, by (S2), read as her subjective valuation augmented by fees and carrying costs, net of the market price, and economically is what one more share shorted would cost her. Intuitively, the sum of these two net costs of going a long, $\tilde{\mu}_t^{P,s}$, and a short, $\tilde{\mu}_t^{P,\theta}$, actually yields “transaction friction” (total wedge) on the short which turns out to be the basic fee and the carrying cost in the paper’s model. However, $\tilde{\mu}_t^{P,s}$ and $\tilde{\mu}_t^{P,\theta}$ are shadow prices on the model’s internal non-negativity constraints, they exist only inside the solved equilibrium. There is no market that quotes them, and no data provider observes such Lagrangian multipliers. Hence, [Proposition 2](#) implies that the “transaction friction” is instead measured by data of wedges, defined on the right side of (34) and (35), which are possibly observable. That is the *implication* of the proposition.

One leg or the other, never both. Nothing forbids the Pessimist from carrying a long and a short at the same time. What rules the redundant pairs out is the frictions, and the total wedge equation (34) is the observation that carrying both legs is offset exactly, but the short still pays the basic fee and carries the cost. That results the following:

Corollary 2 (Pessimist: No long-short coexistence). If $\gamma > 0$, then at every history

$$s_{t+1}^P \theta_{t+1}^P = 0. \tag{36}$$

With $\gamma = 0$, the same conclusion holds at every history at which $p_t^\theta > 0$.

Proof. If $\theta_{t+1}^P = 0$ the product vanishes. If $\theta_{t+1}^P > 0$, then complementary slackness on the short leg’s non-negativity gives $\tilde{\mu}_t^{P,\theta} = 0$, and (34) reduces to $\tilde{\mu}_t^{P,s} = \gamma \theta_{t+1}^P + p_t^\theta$. If $\gamma > 0$, the right-hand side is strictly positive, since $\gamma \theta_{t+1}^P > 0$ and $p_t^\theta \geq 0$. Hence $\tilde{\mu}_t^{P,s} > 0$, and

complementary slackness on the long leg's non-negativity gives $s_{t+1}^P = 0$. When $\gamma = 0$, the same steps go through at any history with $p_t^\theta > 0$, since the right-hand side is then still strictly positive. $\square$

Remark 4 (Why the loan market is not written net of Pessimist's own holdings). The loan market clearing condition (iii) of [Definition 1](#) sets $\ell_{t+1}^O = \theta_{t+1}^P$, and a natural objection is that it should read $\ell_{t+1}^O = \theta_{t+1}^P - s_{t+1}^P$: if the Pessimist owns shares by doing the long, and wants to be net short would borrow only the remainder, so the Optimist needs supply only that difference. The institutional intuition is correct and the model agrees with it. What the model declines to do is impose it. [Corollary 2](#) says $s_{t+1}^P \theta_{t+1}^P = 0$, so wherever she borrows ($\theta_{t+1}^P > 0$), her gross long is zero ($s_{t+1}^P = 0$) and $\theta_{t+1}^P - s_{t+1}^P = \theta_{t+1}^P$: the two conditions coincide at her optimum. While the condition (iii) still holds at no shorting as $\ell_{t+1}^O = \theta_{t+1}^P = 0$.

4.8 Recursive Competitive Equilibrium

To characterize the stationary dynamics of the economy and enable numerical solutions, this subsection cast the sequential competitive equilibrium (SCE, [Definition 1](#)) into a recursive competitive equilibrium (RCE) framework.

From four position variables to one state. The first step is to identify the state that will tractably define the RCE. The four position variables $(s_{t+1}^P, \theta_{t+1}^P, s_{t+1}^O, \ell_{t+1}^O)$ of the SCE are not independent in equilibrium, and one number carries all of them.

Lemma 2 (Net asset position). Define the Pessimist's net position $a_{t+1}^P := s_{t+1}^P - \theta_{t+1}^P$, and write $\text{pos}(x) = \max(0, x)$ and $\text{neg}(x) = \max(0, -x)$. In any sequential competitive equilibrium of [Definition 1](#) with $\gamma > 0$, all four positions are recovered from a_{t+1}^P alone:

$$s_{t+1}^P = \text{pos}(a_{t+1}^P), \quad \theta_{t+1}^P = \text{neg}(a_{t+1}^P), \quad \ell_{t+1}^O = \text{neg}(a_{t+1}^P), \quad s_{t+1}^O = 1 - a_{t+1}^P, \quad (37)$$

and $a_{t+1}^P \in [-\kappa/(1 - \kappa), 1]$.

Proof. Three conditions do the eliminating, one variable each. [Corollary 2](#) of [Section 4.7](#) gives $s_{t+1}^P \theta_{t+1}^P = 0$, so at most one of the Pessimist's two legs is ever positive: her net position therefore carries both, with $a_{t+1}^P > 0$ a gross long and $a_{t+1}^P < 0$ a gross short of that size, which is the first two entries of [\(37\)](#). Loan clearing [\(S13\)](#) sets $\ell_{t+1}^O = \theta_{t+1}^P$, giving the third. Share clearing [\(S12\)](#), $s_{t+1}^P + s_{t+1}^O = 1 + \theta_{t+1}^P$, gives $s_{t+1}^O = 1 + \theta_{t+1}^P - s_{t+1}^P = 1 - a_{t+1}^P$, which is the fourth. For the range, $\theta_{t+1}^P \leq \kappa/(1 - \kappa)$ by [\(13\)](#) places the lower end, and $s_{t+1}^O \geq 0$ places the upper end at $a_{t+1}^P = 1$. $\square$

One state therefore carries the whole cross-section of positions, the lower end of its range being the feasibility wall on the Pessimist's short and the upper end the point at which the Optimist's book is exhausted, $s_{t+1}^O = 0$.

The recursive equilibrium. With the state defined in [Lemma 2](#) ($a^P := s^P - \theta^P$), the equilibrium object can be restated in the form the solver works with. Let z denote the current dividend state, so that $d = d_z$, and let the state be the pair (a^P, z) . What follows is the [equilibrium system \(SCE\)](#) of [Section 4.5](#) rewritten in that state, condition by condition: clause (i) carries [\(S1\)](#) to [\(S4\)](#), clause (ii) carries [\(S5\)](#) and [\(S6\)](#), clause (iii) carries [\(S7\)](#) to [\(S11\)](#), and clause (iv) carries [\(S12\)](#) and [\(S13\)](#). Clause (v) is the one condition with no counterpart there.

Definition 2 (Recursive competitive equilibrium). Given the primitives of [Definition 1](#), a *recursive competitive equilibrium* is a set of functions of the state (a^P, z) , comprising prices $p^s(\cdot)$ and $p^\theta(\cdot)$, consumption policies $c^P(\cdot)$ and $c^O(\cdot)$, a transition rule $\mathcal{A}(\cdot)$, and the five multiplier functions $\tilde{\mu}^{P,s}(\cdot)$, $\tilde{\mu}^{P,\theta}(\cdot)$, $\tilde{\mu}^{O,s}(\cdot)$, $\tilde{\mu}^{O,\ell}(\cdot)$ and $\tilde{\mu}^{O,\text{cap}}(\cdot)$, together with a nonempty compact $S \subseteq [-\kappa/(1 - \kappa), 1]$, such that at every (a^P, z) with $a^P \in S$, writing $a' = \mathcal{A}(a^P, z)$ for the next state, recovering the chosen positions $(s^{P'}, \theta^{P'}, s^{O'}, \ell^{O'})$ from it through [\(37\)](#), and writing

$$V^i = \mathbb{E}_z^i \left[\beta \left(\frac{c^i(a', z')}{c^i(a^P, z)} \right)^{-\sigma} (p^s(a', z') + d_{z'}) \right], \quad i \in \{P, O\},$$

for each agent's continuation value under her own Π^i :

(i) the Euler conditions (S1) to (S4) hold in all four instruments:

$$\begin{aligned}
[s^{P'}] : \quad 0 &= -p^s + V^P + \tilde{\mu}^{P,s} \\
[\theta^{P'}] : \quad 0 &= p^s - p^\theta - \gamma \text{neg}(a') - V^P + \tilde{\mu}^{P,\theta} \\
[s^{O'}] : \quad 0 &= -p^s + V^O + \tilde{\mu}^{O,s} + \kappa \tilde{\mu}^{O,\text{cap}} \\
[\ell^{O'}] : \quad 0 &= p^\theta + \tilde{\mu}^{O,\ell} - \tilde{\mu}^{O,\text{cap}}
\end{aligned} \tag{38}$$

(ii) the budget constraints (S5) and (S6) hold:

$$\begin{aligned}
c^P + p^s a' + p^\theta \text{neg}(a') + \frac{1}{2} \gamma (\text{neg}(a'))^2 &= (p^s + d_z) a^P + Y^P \\
c^O + p^s (1 - a') - p^\theta \text{neg}(a') - \chi \frac{1}{2} \gamma (\text{neg}(a'))^2 &= (p^s + d_z) (1 - a^P) + Y^O
\end{aligned} \tag{39}$$

(iii) the complementary slackness conditions (S7) to (S11) hold:

$$\begin{aligned}
[s^{P'}] : \quad \tilde{\mu}^{P,s} &\geq 0, & \text{pos}(a') &\geq 0, & \tilde{\mu}^{P,s} \text{pos}(a') &= 0 \\
[\theta^{P'}] : \quad \tilde{\mu}^{P,\theta} &\geq 0, & \text{neg}(a') &\geq 0, & \tilde{\mu}^{P,\theta} \text{neg}(a') &= 0 \\
[s^{O'}] : \quad \tilde{\mu}^{O,s} &\geq 0, & 1 - a' &\geq 0, & \tilde{\mu}^{O,s} (1 - a') &= 0 \\
[\ell^{O'}] : \quad \tilde{\mu}^{O,\ell} &\geq 0, & \text{neg}(a') &\geq 0, & \tilde{\mu}^{O,\ell} \text{neg}(a') &= 0 \\
[\kappa s^{O'} - \ell^{O'}] : \quad \tilde{\mu}^{O,\text{cap}} &\geq 0, & \kappa (1 - a') - \text{neg}(a') &\geq 0, & \tilde{\mu}^{O,\text{cap}} (\kappa (1 - a') - \text{neg}(a')) &= 0
\end{aligned} \tag{40}$$

where non-negativity restrictions on $s^{P'}$, $\theta^{P'}$ and $\ell^{O'}$ are satisfied since pos and neg are non-negative by construction; $s^{O'} \geq 0$ is the upper endpoint $a' \leq 1$; and the cap $\kappa s^{O'} - \ell^{O'} \geq 0$ is the lower endpoint $a' \geq -\kappa/(1 - \kappa)$, which is the feasibility wall (13).

(iv) the market clearing conditions (S12) and (S13):

$$\begin{aligned}
[\text{shares}] : \quad & \underbrace{\text{pos}(a') + (1 - a')}_{s^{P'} + s^{O'}} = 1 + \underbrace{\text{neg}(a')}_{\theta^{P'}} \\
[\text{loans}] : \quad & \underbrace{\text{neg}(a')}_{\ell^{O'}} = \underbrace{\text{neg}(a')}_{\theta^{P'}}
\end{aligned} \tag{41}$$

holds automatically by $\text{pos}(x) - x = \text{neg}(x)$ for every x , clearing is therefore imposed by the state reduction rather than by an equation the solver must satisfy; and:

(v) S is invariant under the transition: $\mathcal{A}(a^P, z) \in S$ for every $a^P \in S$ and every z .

Clause (v) is the one that does not appear in Definition 1 and the one that is usually left implicit. A sequential equilibrium is a path; a recursive equilibrium is a fixed set of functions on a fixed domain, and it exists only if the domain is one the policy does not leave.

4.9 The Solving System (RCE)

Follows Definition 2, this subsection assembles a state-reduced RCE system for the solver.

First. Where the Pessimist is short, $a' < 0$, her short leg is interior, so the complementary slackness (40) sets $\tilde{\mu}^{P,\theta} = 0$ and the $[\theta^P]$ row (38) reads $0 = p^s - p^\theta - \gamma \text{neg}(a') - V^P$. Where she is long, $a' > 0$, her long leg is interior, so $\tilde{\mu}^{P,s} = 0$ and the $[s^P]$ row reads $0 = -p^s + V^P$. One equation carries both cases,

$$0 = -p^s - (p^\theta + \gamma \text{neg}(a')) \text{neg}'(a') + V^P, \tag{42}$$

since $\text{neg}'(a') = -1$ where she is short, which returns the first, and $\text{neg}'(a') = 0$ where she is long, which returns the second.

Second. Both multipliers $\tilde{\mu}^{P,s}, \tilde{\mu}^{P,\theta}$ dropped. By Corollary 2, the Pessimist exhibits mutually exclusive long and short positions. So in each region (42) reproduces the active leg's row, whose multiplier vanishes by complementary slackness. The idle leg's row still holds,

but it now only pins that leg's multiplier as a residual, and since neither of the Pessimist's multipliers is an unknown of the reduced system, the row is dropped.

Finally. What left from the clauses (i) to (iv) of [Definition 2](#) are eight equations in eight unknowns: the two prices p^s and p^θ , the two consumptions c^P and c^O , the state's image a' , and the Optimist's three multipliers $\tilde{\mu}^{O,s}$, $\tilde{\mu}^{O,\ell}$ and $\tilde{\mu}^{O,\text{cap}}$. Writing $x \equiv \kappa(1 - a') - \text{neg}(a')$ for the cap's slack, the eight are:

$$-p^s - \gamma \text{neg}(a') \text{neg}'(a') - p^\theta \text{neg}'(a') + V^P = 0 \quad (\text{R1})$$

$$-(p^s - \kappa \tilde{\mu}^{O,\text{cap}}) + V^O + \tilde{\mu}^{O,s} = 0 \quad (\text{R2})$$

$$p^\theta + \tilde{\mu}^{O,\ell} - \tilde{\mu}^{O,\text{cap}} = 0 \quad (\text{R3})$$

$$(p^s + d_z)a^P + Y^P - p^s a' - p^\theta \text{neg}(a') - \frac{1}{2}\gamma(\text{neg}(a'))^2 - c^P = 0 \quad (\text{R4})$$

$$(p^s + d_z)(1 - a^P) + Y^O - p^s(1 - a') + p^\theta \text{neg}(a') + \chi \frac{1}{2}\gamma(\text{neg}(a'))^2 - c^O = 0 \quad (\text{R5})$$

$$\tilde{\mu}^{O,s} \geq 0, \quad 1 - a' \geq 0, \quad \tilde{\mu}^{O,s}(1 - a') = 0 \quad (\text{R6})$$

$$\tilde{\mu}^{O,\ell} \geq 0, \quad \text{neg}(a') \geq 0, \quad \tilde{\mu}^{O,\ell} \text{neg}(a') = 0 \quad (\text{R7})$$

$$\tilde{\mu}^{O,\text{cap}} \geq 0, \quad x \geq 0, \quad \tilde{\mu}^{O,\text{cap}} x = 0 \quad (\text{R8})$$

(R1) is the Pessimist's collapsed pair (42). (R2) and (R3) are the $[s^O]$ and $[\ell^O]$ rows of clause (i) unchanged. (R4) and (R5) the two budgets of clause (ii). (R6) to (R8) the three complementarity conditions of clause (iii) that survive the reduction. Clause (iv) holds automatically by construction.

5 Quantitative and Numerical Implementation

This section outlines the model’s global numerical implementation, detailing parameter calibration, the subjective dividend chain, and the solution grid. We formalize how the global solution algorithm addresses non-differentiability at strategy kinks, enforces complementarity conditions for the capacity constraints. The non-linear equilibrium system is solved globally via the Policy Function Iteration method using the [GDSGE](#) toolbox developed by [Cao, Luo, and Nie \(2023\)](#), as detailed in [Appendix C](#).

5.1 Parameterization and Subjective Belief Heterogeneity

Baseline preferences, endowments, and lending frictions and other parameters are calibrated and summarized in [Table 2](#).

Symbol	Meaning	Value
β	Discount factor, common	0.95
σ	Relative risk aversion, common	1.5
Y	Labour endowment, common to both agents	0.5
κ	Lendable fraction of a lender’s book	0.5
γ	Carrying-cost coefficient	0.30
χ	Share of the carrying cost the lender receives	0.40
ρ	Dividend autocorrelation	0.9
σ_ε	Dividend innovation standard deviation	0.01
N	Dividend states	5
ε	Corner-smoothing parameter	10^{-4}
α	Penalty stiffness for the cap	10

Table 2: Calibration Summary. The parameters are grouped into preference and endowments, lending frictions, and the dividend process. The final two rows represent purely computational parameters required for numerical regularization.

Preference and Endowment Parameters. The preference and endowment blocks follow standard asset-pricing conventions. The common subjective discount factor is set to $\beta = 0.95$, and the coefficient of relative risk aversion is fixed at $\sigma = 1.5$. Both agents receive symmetric,

time-invariant labor endowments normalized to $Y = 0.5$.

Temporal Interpretation and the Model Clock. The model does not impose a fixed calendar frequency on a single model period. While the subjective discount factor $\beta = 0.95$ and dividend autocorrelation $\rho = 0.9$ match standard numerical values in the low-frequency macro-finance literature, they are used here to discipline the model’s internal velocity rather than to freeze time into annual blocks. The model therefore operates on an internal “model clock”, and simulation horizons and event-time axes measure the duration of shorting episodes in model periods rather than imposing a one-to-one mapping to calendar days. Model fee levels are reported per model period and are not annualized. The main quantitative comparisons in the paper are ratios, which are invariant to the assumed period length.

Lending Market Frictions. The friction block introduces three parameter values to discipline the securities lending channel. The lendable fraction constraint is restricted to $\kappa = 0.5$, which parameterizes the upper bound on feasible allocations derived in [Lemma 1](#) and the lower bound in [Lemma 2](#). The institutional intermediary split is fixed at $\chi = 0.40$, reflecting the empirical reality that beneficial owners typically retain 30% to 50% of transacted borrow fees. Finally, the carrying-cost elasticity parameter is set to $\gamma = 0.30$. As documented in [Appendix D.1](#), a numerical solution to the global competitive equilibrium exists only within the compact subset $\gamma \in [0.10, 0.50]$, positioning our baseline at the structural midpoint. As [Table D.3](#) demonstrates, this baseline balances a key empirical conflict: a higher elasticity pushes the simulated fee ramp closer to the data but carries asset returns further away. A standard point-fit calibration is unfeasible here because this trade-off is bounded; even at the maximum permissible value ($\gamma = 0.50$), the model’s peak fee ramp remains well below the empirical benchmark.

Dividend Discretization and Grid Construction. The exogenous objective dividend process is governed by a continuous AR(1) specification with persistence $\rho = 0.9$ and

innovation standard deviation $\sigma_\varepsilon = 0.01$. We discretize this process into a finite state Markov chain with $N = 5$ states using the method of [Tauchen \(1986\)](#), which yields the symmetric support vector:

$$d \in \{0.1176, 0.1463, 0.1750, 0.2037, 0.2324\}.$$

To adequately capture tail behavior, the discrete grid is centered at the unconditional mean $\bar{d} = 0.175$ and spans ± 2.5 unconditional standard deviations, σ_d . Due to fundamental persistence, the scaling factor expands by $1/\sqrt{1 - \rho^2} = 2.29$:

$$\sigma_d = \frac{\sigma_\varepsilon}{\sqrt{1 - \rho^2}} = 0.0229, \quad \text{where } d = 0.175 \pm \{0, 1.25, 2.5\} \sigma_d.$$

The grid nodes are spaced uniformly at intervals of $1.25\sigma_d$.

Dividend Persistence and Lending Friction Interaction. The choice of a high dividend persistence ($\rho = 0.9$) matches empirical macro-finance benchmarks and ensures that fundamental shocks generate long-lived conditional valuation wedges between the agents. Without this temporal persistence, shocks would mean-revert too rapidly, preventing short positions from expanding sufficiently to interact with the lending frictions. In our quantitative simulations, this expansion is smoothly moderated by the convex carrying cost, which operates as the active clearing mechanism while the physical inventory cap constraint (κ) remains strictly slack. Because the carrying cost is strictly quadratic in position size, the marginal cost of shorting escalates sharply with volume, acting as an endogenous economic brake. The Pessimist voluntarily restricts her short position at an optimal threshold where marginal benefits equal marginal costs, peaking well before depleting the available borrowable float. Consequently, the model clears through the smooth price-slope margin (γ) rather than hitting a hard, non-differentiable quantity rationing boundary, providing a micro-founded rationalization for why borrow fees can spike dramatically even when borrowable inventory is not entirely exhausted.

Subjective belief heterogeneity. Belief heterogeneity is operationalized by assigning the two agents distinct long-run unconditional means over an identical state space. The Pessimist’s transition matrix, Π^P , represents the objective data-generating process: it is the symmetric Tauchen discretization centered at the true structural mean $\mu^P = 0.1750$, as defined in (43). The economy is simulated entirely under the objective measure Π^P , rendering the short-selling Pessimist the fundamentally informed agent and isolating the model’s asset mispricing within the Optimist’s subjective distortion.

The Optimist’s transition matrix, Π^O , is constructed over the same five grid points using identical persistence ($\rho = 0.9$) and innovation volatility ($\sigma_\varepsilon = 0.01$), but parameterizes a subjective process reverting to a higher unconditional mean, $\mu^O = 0.2188$. Maintaining a shared, time-invariant grid ensures the disagreement is well-posed: both agents attach differing subjective probabilities to an identical support rather than evaluating divergent state spaces. This structural design models disagreement over the long-run level of fundamentals rather than short-run transitory shocks, allowing the speculative positions to persist across long horizons before market frictions clear them.

Because the state space is centered around the Pessimist’s mean, the Optimist’s subjective transition matrix exhibits strong structural asymmetry, as shown in (44). High dividend states are significantly more persistent under her measure ($[\Pi^O]_{55} = 0.9029$) than low states ($[\Pi^O]_{11} = 0.6636$), meaning the Optimist implicitly views positive shocks as both more frequent and structurally stickier.

Symmetric objective beliefs (Pessimist, Π^P):

$$\Pi^P = \begin{pmatrix} 0.805191 & 0.194713 & 0.000096 & 0.000000 & 0.000000 \\ 0.042660 & 0.831666 & 0.125645 & 0.000030 & 0.000000 \\ 0.000008 & 0.075799 & 0.848385 & 0.075799 & 0.000008 \\ 0.000000 & 0.000030 & 0.125645 & 0.831666 & 0.042660 \\ 0.000000 & 0.000000 & 0.000096 & 0.194713 & 0.805191 \end{pmatrix} \quad (43)$$

Asymmetric subjective beliefs (Optimist, Π^O):

$$\Pi^O = \begin{pmatrix} 0.663601 & 0.335898 & 0.000501 & 0.000000 & 0.000000 \\ 0.015440 & 0.745423 & 0.238963 & 0.000174 & 0.000000 \\ 0.000001 & 0.030613 & 0.809725 & 0.159606 & 0.000056 \\ 0.000000 & 0.000004 & 0.056470 & 0.843713 & 0.099813 \\ 0.000000 & 0.000000 & 0.000015 & 0.097075 & 0.902910 \end{pmatrix} \quad (44)$$

5.2 Corner Smoothing

The [equilibrium system \(RCE\)](#) carries one state instead of four to solve the model, and that all four positions are recovered from a' ($\equiv a_{t+1}^P$) through `pos` and `negp`, neither of which is differentiable at zero.

That non-differentiability enters the solved system directly (i.e. GDSGE), because the Pessimist's Euler equation carries both $\theta' = \text{neg}(a')$ and the derivative $\text{neg}'(a')$. The function is therefore smoothed:

$$\begin{aligned} \text{neg}_\varepsilon(x) &= \frac{1}{2} \left(\sqrt{x^2 + \varepsilon^2} - x \right) \longrightarrow \max(0, -x) = \text{neg}(x) \quad \text{as } \varepsilon \rightarrow 0 \\ \text{pos}_\varepsilon(x) &= \frac{1}{2} \left(\sqrt{x^2 + \varepsilon^2} + x \right) \longrightarrow \max(0, x) = \text{pos}(x) \quad \text{as } \varepsilon \rightarrow 0 \end{aligned} \quad (45)$$

both differentiable everywhere, which is what gives the inner Newton step a defined derivative at the kink the solution occupies.

Differentiating (45) gives:

$$\text{neg}'_\varepsilon(x) = \frac{1}{2} \left(\frac{x}{\sqrt{x^2 + \varepsilon^2}} - 1 \right), \quad \text{pos}'_\varepsilon(x) = \frac{1}{2} \left(\frac{x}{\sqrt{x^2 + \varepsilon^2}} + 1 \right).$$

Smoothing causes minor accounting errors, but [Appendix D.5](#) shows they are negligibly small ($< 10^{-6}$ grid-wide, $< 10^{-8}$ ergodically).

5.3 Imposing Complementarity Conditions

This technique is well-done but not necessary for the baseline model with both the base fee and the premium carrying cost, because its solved policy already stays inside the feasibility wall of [Lemma 2](#), which is the invariance column of [Table D.4](#). It is needed for the cases that converge to a policy outside that wall, as the frictionless benchmark of [Figure F.1](#) does.

The complementary slackness (CS) conditions of the [equilibrium system \(RCE\)](#) can be numerically implemented as follows.

Enforcing Optimist non-negativity. Consider the CS in [\(R6\)](#):

$$\tilde{\mu}^{O,s} \geq 0, \quad 1 - a' \geq 0, \quad \tilde{\mu}^{O,s} (1 - a') = 0.$$

The Optimist’s long position, $s^O = 1 - a'$, is non-negative given the upper bound $a' \leq 1$ of the state range, which [Lemma 2](#) establishes. To analyze the boundary “leak” and resulting non-existence of equilibrium (i.e., that happens in the *frictionless* shorting economy as detailed in [Appendix F](#)), the solver must evaluate convergence behaviors where $a' > 1$ and therefore $1 - a' < 0$. In these negative regions, we map the position using the positive projection $s^{O'} = \text{pos}(1 - a')$.

The Penalty-Based Lending Constraint Approximation. The CS in [\(R8\)](#):

$$\tilde{\mu}^{O,\text{cap}} \geq 0, \quad x \geq 0, \quad \tilde{\mu}^{O,\text{cap}} x = 0,$$

where $x = \kappa(1 - a') - \text{neg}(a') \equiv \kappa s^{O'} - \ell^{O'}$, cannot be written the same way. The other two conditions are exact as products because both of their signs are already imposed: the multipliers are bounded below at zero in the solver, and the slacks are the projections pos and neg , which are non-negative by construction. Here the slack is a difference of two non-negative quantities and nothing bounds it below, so the product $\tilde{\mu}^{O,\text{cap}} x$ would also be satisfied by

breaching the cap, $x < 0$, at zero shadow price. We therefore replace it with a single penalty equation:

$$\tilde{\mu}^{O,\text{cap}} - \alpha \text{neg}(x) = 0. \quad (46)$$

Under the penalty formulation in (46), the complementarity condition fails when the lending capacity is breached, $x < 0$. Instead of strictly forbidding over-capacity states, the penalty function returns a positive multiplier $\tilde{\mu}^{O,\text{cap}} = -\alpha x > 0$, yielding a negative product $-\alpha x^2 < 0$ that violates the standard complementary slackness conditions in (R8). Rather than enforcing a hard boundary, the penalty method charges a cost for crossing the feasibility wall (13), $a' \geq -\kappa/(1 - \kappa)$. As the stiffness parameter $\alpha \rightarrow \infty$, this soft charge steepens, compressing the tolerated violation zone toward a true hard cap. This formulation allows the shadow price of lending capacity to scale with the capacity shortfall. Therefore, the penalty is what makes the basic fee a function of the state at all, and is especially useful to analyze the boundary “leak” and resulting non-existence of equilibrium.

5.4 Grid Infrastructure and Global Solution Methodology

To capture the model’s severe non-linearities, including asset pricing spikes, lending capacity limits, and policy function non-differentiabilities where the Pessimist’s optimal portfolio transitions between long and short regimes, we solve the system globally using computational projection methods via the GDSGE toolbox.

State-Space Discretization and Endogenous Bounds. The continuous endogenous state variable tracking the Pessimist’s net equity position, a^P , is discretized onto a dense, uniformly spaced grid comprising 200 nodes bounded within the interval $[-1, 1]$. Analytical validation via Lemma 2 confirms that under the baseline calibration of $\kappa = 0.5$, the structural feasibility set is exactly $[-\kappa/(1 - \kappa), 1] = [-1, 1]$. Expanding the state space beyond this lower boundary would venture into unfeasible allocations that the securities lending market cannot structurally support. Crucially, the solver’s policy choice boundaries are parameterized

strictly wider than the collocation grid. This structural design ensures that any candidate policy function attempting to breach the admissible state space is explicitly captured and evaluated rather than artificially clipped, rendering clause (v) of the Competitive Equilibrium definition ([Definition 2](#)) numerically verifiable.

Algorithmic Convergence and Policy Iteration. The equilibrium system is solved globally by policy function iteration in the GDSGE toolbox. Each sweep solves the eight equations of [Section 4.9](#) at every node of the state grid, then refreshes the three functions the next sweep needs: the two consumptions and the share price. These are the only quantities the system requires dated $t + 1$, and all three enter through the continuation values V^P and V^O of [Definition 2](#)—the consumptions in the marginal-utility ratio, the price in the share’s payoff. They alone are interpolated and carried across iterations, so the fixed point is a fixed point in them. Iteration stops when the largest change in any of the three, at any point of the grid, falls below 10^{-6} (`TolEq = 1e-6`):

$$\max_{f \in \{c^P, c^O, p^s\}} \sup_{a^P, z} |f_{n+1}(a^P, z) - f_n(a^P, z)| < 10^{-6},$$

the supremum running over the entire tensor product of the state grid. The remaining five unknowns—the state’s image a' , the fee p^θ , and the Optimist’s three multipliers—never appear dated $t + 1$; they are recovered within each sweep by the inner solve at every node, and their accuracy is governed by `TolSol` rather than `TolEq`. [Appendix C](#) sets out the full algorithm.

6 The Equilibrium Across the State Space

This section characterizes the solved equilibrium as a function of the state, before the next sets it in motion. The policy and pricing rules are defined over the whole feasible set, but the economy visits only a small part of it, so the discussion opens with the region the stationary distribution occupies and reads what follows within it. The remaining subsections take the equilibrium apart in turn: the portfolio allocations and the lending capacity they draw on, the asset price and the wedges that separate it from the Pessimist's own valuation, the utilization of lendable inventory against its ceiling, and the restoring force that makes the configuration stable.

6.1 Equilibrium Existence and the Ergodic Footprint

Panel (a) of [Figure 3](#) plots the globally solved transition rule across the entire support of the state space. The endogenous policy mapping yields an image which lies strictly within the interior of the grid boundaries. This validates that the policy maps the state space into itself, satisfying the structural invariance required by clause (v) of the Competitive Equilibrium framework ([Definition 2](#)).

Projecting the joint probability distribution of the endogenous state and the exogenous Markov dividend chain to its fixed point yields the long-run stationary distribution illustrated in panel (b). The ergodic mass is tightly concentrated within a compact region:

$$a_t^P \in [-0.120, 0.033],$$

occupying less than 8% of the total transition grid, while over the remainder of the domain the density is numerically zero. The distribution's mode is located at $a_t^P = -0.0452$. Within this localized footprint, the Pessimist operates as an active short-seller on 97.54% of simulated dates, holding her long asset position at its corner boundary ($s_{t+1}^P = 0$) while her optimal

short position varies dynamically in the interior. On the remaining 2.46% of dates, the net asset state turns positive, causing the Pessimist to hold a small long position and completely deactivating the lending market friction. Notably, because the economy never approaches the physical boundary walls, the simulated market frictions are driven entirely by the policy rule near the ergodic band.

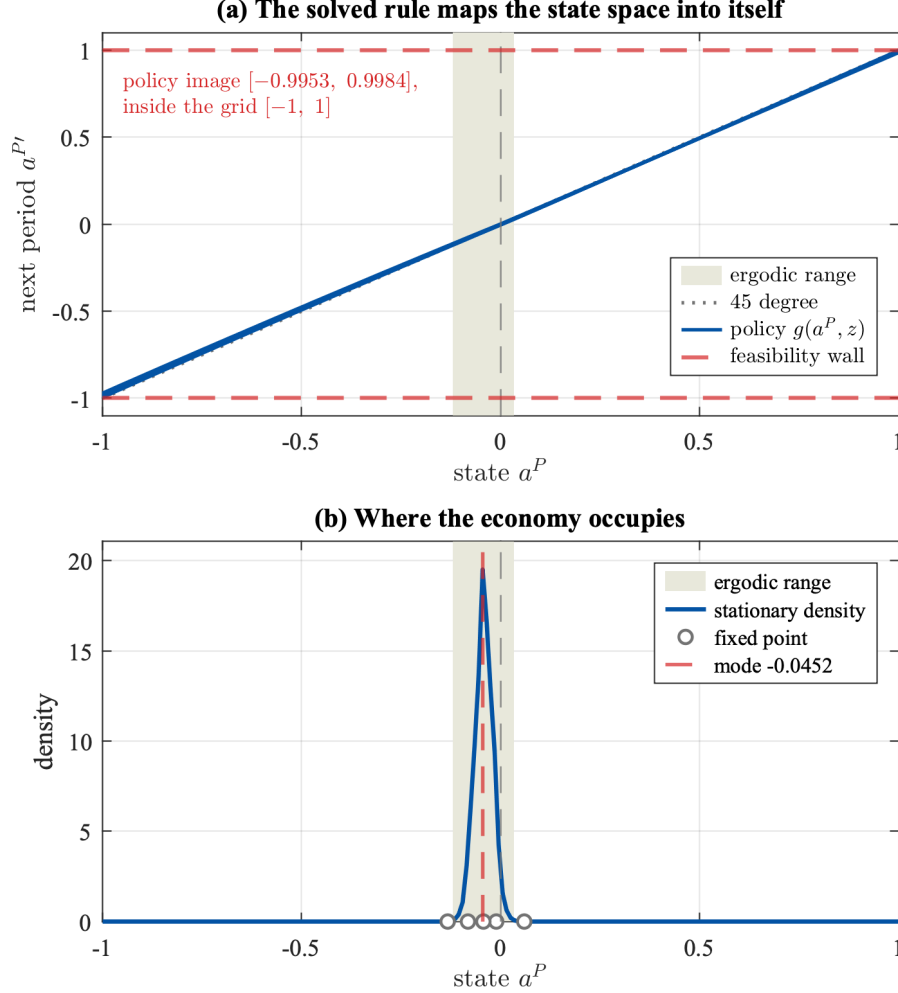


Figure 3: Equilibrium Existence and the Ergodic Footprint. Both panels share an identical horizontal axis representing the full feasible set. The dashed vertical line marks the portfolio regime threshold $a_t^P = 0$, where the Pessimist switches between asset holding and short-selling. *Panel (a):* The solved transition policy $g(a_t^P, z_t)$ across all five dividend states, plotted alongside the 45-degree line (dotted) and the physical feasibility walls $[-\kappa/(1-\kappa), 1]$ (dashed). *Panel (b):* The ergodic stationary density of the state variable. The shaded region denotes the compact ergodic range, and the circles along the horizontal axis identify the five conditional fixed points detailed in Table 3. Source: `gmod/fee_cost/plot_ergodic_footprint.m`.

d	Weight $w(z)$	Fixed Point	Local Slope
0.1176	0.0535	+0.0593	0.980
0.1463	0.2442	−0.0111	0.946
0.1750	0.4047	−0.0434	0.955
0.2037	0.2442	−0.0821	0.963
0.2324	0.0535	−0.1331	0.971

Table 3: Conditional Fixed Points and Ergodic Properties. Each exogenous dividend state dictates a unique, locally stable conditional attractor where the policy rule intersects the 45-degree line. Because all local slopes are strictly less than unity, each state is locally contracting. Only the lowest dividend state features an attractor on the long side of the state space ($a_t^P > 0$), explaining why the Pessimist occasionally holds long positions. Bold text highlights the median state under the symmetric objective measure Π^P ; its conditional fixed point (−0.0434) lies immediately adjacent to the aggregate ergodic mode (−0.0452). Source: `gmod/fee_cost/plot_ergodic_footprint.m`.

6.2 Portfolio Allocations and Lending Capacity Restrictions

Figure 4 illustrates the two agents’ equilibrium portfolio allocations and the endogenous lending capacity bounds as functions of the state variable a_t^P (the Pessimist’s net equity position, $s_t^P - \theta_t^P$). The agents’ distinct market roles emerge endogenously within this framework, mapping out how the securities-lending friction constrains allocation space. The simulated ergodic band derived in Section 6.1 straddles this transition, operating almost entirely within the negative, short-selling regime.

Panel (a) demonstrates a strict specialization property in the Pessimist’s portfolio choice: she never simultaneously maintains long and short equity legs. For $a_t^P \geq 0$, the equity asset clears at her own subjective valuation; consequently, no fundamental mispricing exists for her to exploit, keeping her optimal short position θ_{t+1}^P flat at zero. In this region, she holds gross long shares only, scaling linearly to unity at the upper bound of the state space.

Conversely, when $a_t^P < 0$, this portfolio structure reverses. The Optimist holds the dominant asset position, and her optimistic beliefs support an equilibrium clearing price that strictly exceeds the Pessimist’s reservation value. Under these conditions, the Pessimist holds no long inventory ($s_{t+1}^P = 0$) and actively short-sells the asset, expanding her short position as the valuation wedge widens toward the lower physical feasibility wall at $a_t^P = -1$.

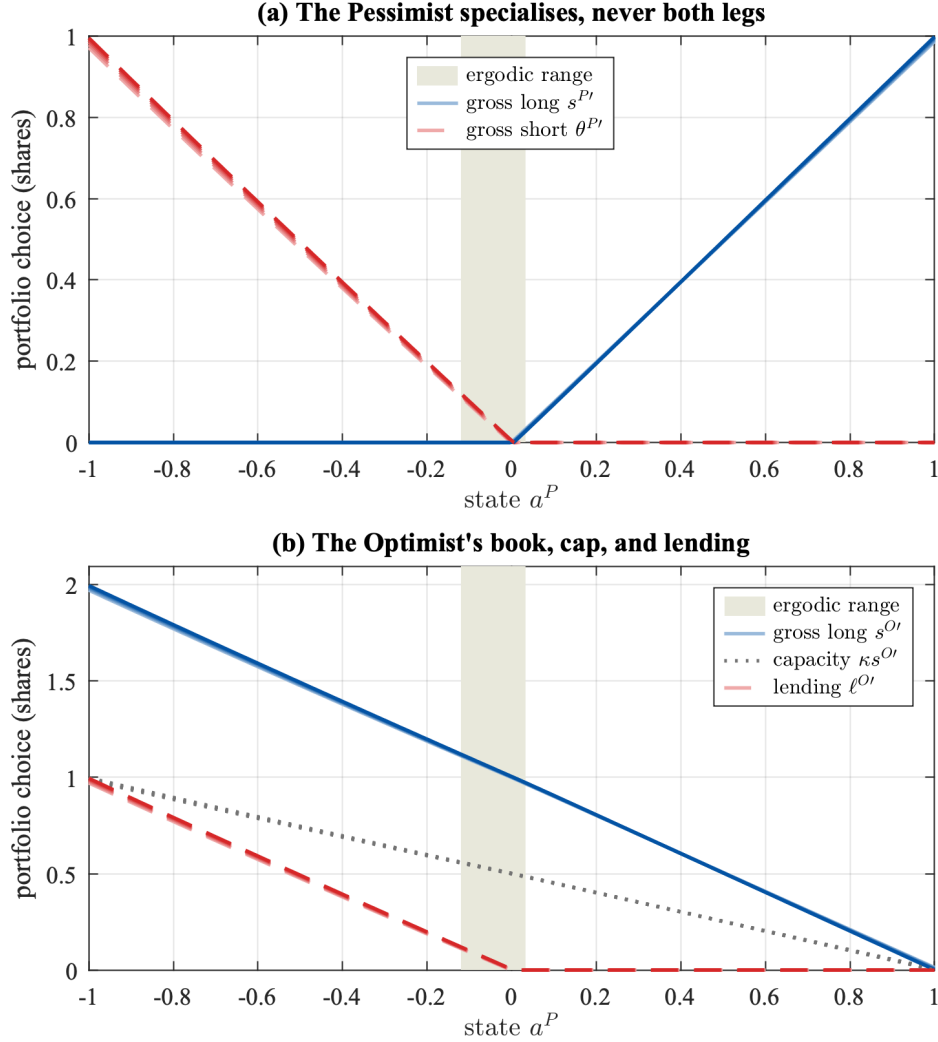


Figure 4: Equilibrium Market Portfolios and Share Lending Capacity. *Panel (a):* The Pessimist's optimal portfolio allocation, demonstrating strict asset-role specialization: she maintains a gross long position (s_{t+1}^P) when her net asset state is positive ($a_t^P > 0$) and switches exclusively to gross short positions (θ_{t+1}^P) when the net state turns negative ($a_t^P < 0$). *Panel (b):* The Optimist's gross allocations and lending capacity. As the Pessimist shorts heavily ($a_t^P \rightarrow -1$), the Optimist absorbs the shorted float, driving her gross long position (solid blue line) above the unitary outstanding supply. Concurrently, the margin between the Optimist's actual share lending (ℓ_{t+1}^O , dashed red line) and her physical capacity constraint (κs_{t+1}^O , dotted line) narrows monotonically as short interest deepens. Source: `gmod/fee.cost/plot.portfolios.capacity.m`.

Panel (b) highlights the corresponding supply-side dynamics. The Optimist absorbs the entire shorted volume issued by the Pessimist, driving her gross long equity position well above the economy’s aggregate asset supply, approaching nearly two full shares near the boundary wall. Because any active short interest must be physically borrowed from long supply, the Optimist’s share lending is structurally capped by the capacity boundary κs_{t+1}^O . The distance between this capacity ceiling and the realized lending volume (ℓ_{t+1}^O) is highly asymmetric around zero, narrowing monotonically as the short interest deepens. [Section 6.3](#) maps these equilibrium quantity allocations into their corresponding price and borrow-fee dynamics.

6.3 Equilibrium Asset Prices and Two-Wedge Frictions

Equity prices and asset allocations are jointly determined components of a single global equilibrium mapping. [Figure 5](#) illustrates this pricing solution, with panel (a) mapping the endogenous equity price p_t^s across the discrete dividend states, and panel (b) decomposing the total per-share cost of shorting into its structural components.

First-Order Conditions and Valuation Wedges. Recall that V_t^i denotes agent i ’s subjective marginal valuation of the equity asset, representing the expected continuation payoff discounted by her respective stochastic discount factor. In the global solution, the Optimist’s equity Euler equation ([R2](#)) holds with an interior, slack multiplier, pinning the market price directly to her subjective valuation across the entire support: $p_t^s = V_t^O$.

By the strict specialization property ([Corollary 2](#)), the Pessimist’s optimal portfolio regimes are mutually exclusive. In regions where she holds long positions ($a_t^P \geq 0$), the marginal constraint yields $\text{neg}'(a_{t+1}) = \max\{0, -a_{t+1}\}' = 0$, reducing her first-order condition ([R1](#)) to $p_t^s = V_t^P$. Trading thus equalizes marginal valuations, and belief heterogeneity manifests as asymmetric portfolio concentration rather than an asset pricing wedge. Conversely, when the Pessimist operates in the short-selling regime ($a_t^P < 0$), an endogenous valuation wedge opens,

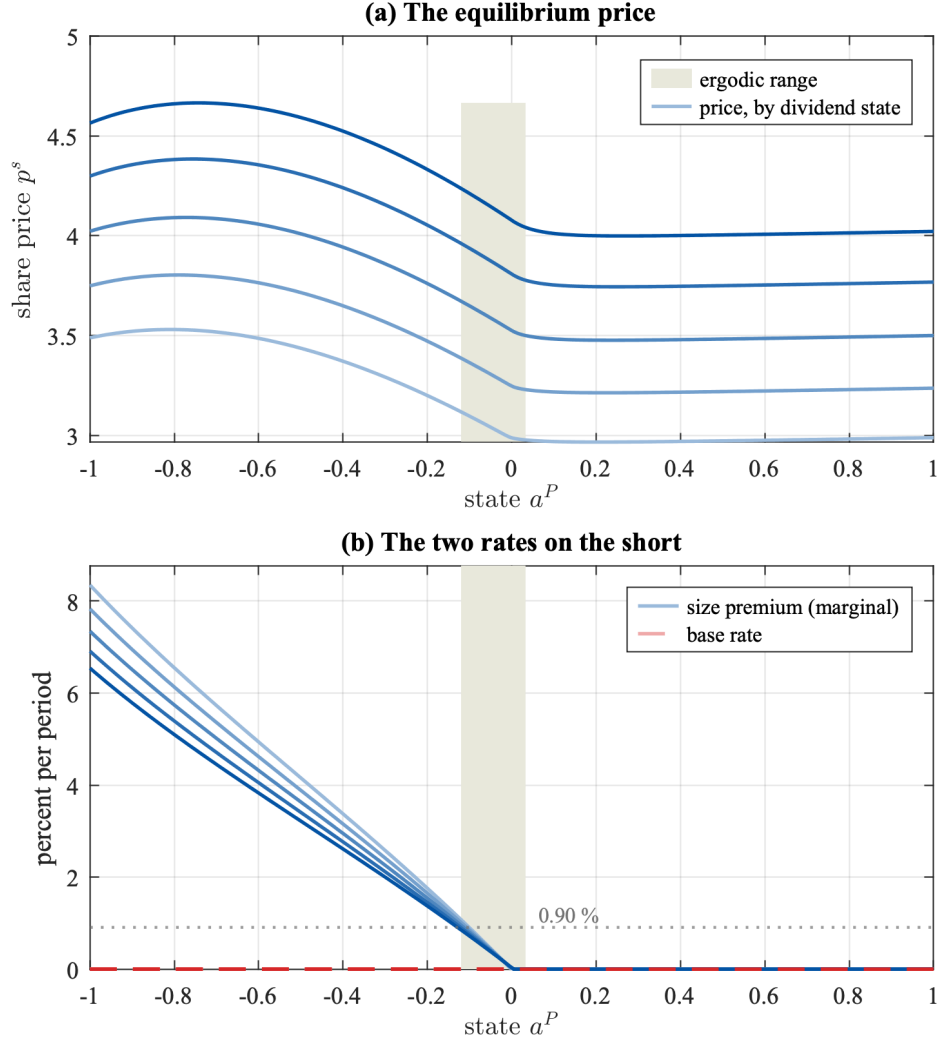


Figure 5: Equilibrium Share Prices and Shorting Frictions. *Panel (a):* The equilibrium asset price p_t^s across the five dividend states (darker lines denote higher dividend states). The pricing functions are invariant to changes in the portfolio state variable to the right of $a_t^P = 0$, but rise monotonically to the left as market clearing shifts equity share concentration toward the Optimist and the clearing price weights her valuation more heavily. The shaded region denotes the simulated ergodic band. *Panel (b):* The two per-share rates on the short. The base rate is numerically zero throughout, so the whole charge is the size premium; a higher dividend carries a lower premium. Source: `gmod/fee_cost/plot_price_frictions.m`.

reflecting the fundamental disagreement: $p_t^s - V_t^P = V_t^O - V_t^P$. Evaluating the portfolio derivative $\partial\theta_{t+1}^P/\partial a_{t+1} = -1$ within this negative region, the Pessimist's short-sale first-order condition simplifies to:

$$p_t^s = V_t^P + \gamma \theta_{t+1}^P + p_t^\theta \iff \gamma \theta_{t+1}^P = (p_t^s - V_t^P) - p_t^\theta. \quad (47)$$

Equation (47) dictates that the endogenous valuation wedge is bounded by the marginal transaction cost of carrying the position: the marginal carrying cost of the final shorted unit exactly matches the subjective overvaluation it exploits, net of the base borrow fee.

Asset Prices Across the Portfolio Regimes. The asset price profiles in panel (a) are structurally partitioned by the zero-short boundary. In the long-holding regime ($a_t^P \geq 0$), equity prices remain flat at their steady-state baselines because both agents maintain unconstrained long positions (slack capacity limit κ , no shorting cost, pure interior optimization), ensuring asset valuations converge and no pricing wedge emerges.

In the short-selling regime ($a_t^P < 0$), equity prices paradoxically rise rather than fall. Asset market clearing requires $s_{t+1}^O = 1 - a_t^P$, forcing the Optimist to absorb an increasing share of the aggregate float. Because the clearing price reflects her valuation throughout this region, the price expands, widening the fundamental valuation gap $p_t^s - V_t^P$. By (47), the Pessimist's optimal short interest scales linearly with this gap, divided by the cost parameter γ .

This asset price expansion is bounded. Because, while the valuation wedge $\gamma\theta_{t+1}^P$ scales linearly with position size, the charge the Pessimist funds out of consumption is strictly quadratic: $\frac{1}{2}\gamma(\theta_{t+1}^P)^2$. In deep shorting states, this quadratic cost compresses the Pessimist's residual consumption, driving up her marginal utility and depressing her subjective valuation of future continuation claims (V_t^P). Past from the peak to the wall, the quadratic term dominates, V_t^P gives up more than the wedge gains. Consequently, the asset price function, $p_t^s = V_t^P + \gamma\theta_{t+1}^P + p_t^\theta$, peaks and turns downward near the boundary wall. As demonstrated

in [Section 6.4](#), this peak lies entirely outside the simulated ergodic band and is never reached along any equilibrium dividend trajectory.

Decomposition of the Endogenous Borrowing Cost. Panel (b) plots the two normalized components of the per-share shorting cost: the base rate of borrow fee, p_t^θ/p_t^s , and the marginal size premium, $\gamma\theta_{t+1}^P/p_t^s$. The base rate (dashed red line) is strictly zero across the entire state space, confirming that available borrowable inventory remains non-binding. The entire borrowing friction is therefore driven by the marginal size premium, which remains zero for $a_t^P \geq 0$ and escalates near-linearly in the negative regime, peaking between 6.5% and 8.4% per period.

This size premium is countercyclical, peaking in the lowest dividend states where the equity price in the denominator is smallest. Economically, this ordering emerges because a positive dividend shock acts as a persistent shift ($\rho = 0.9$), lifting the aggregate expected dividend stream and the underlying stock price. However, because the size premium reflects the scaled disagreement regarding the growth path rather than the baseline level of cash flows, the absolute valuation wedge remains invariant to the shock, causing the normalized per-share rate to compress as prices rise.

Level versus Variation. The model characterizes the state-dependent, dynamic component of the borrowing friction rather than its structural baseline level. The average baseline cost of carrying a hard-to-borrow stock (i.e., 16.5% APR level at $t - 60$ in the event sample) reflects institutional search frictions, inventory costs, and prime broker margins which are out of the model's scope. Accordingly, our framework rationalizes the higher-frequency variations and spikes around this empirical baseline rather than the baseline level itself.

6.4 Capacity Utilization and the Unreached Ceiling

To isolate the structural mechanics underlying the shorting frictions discussed in [Section 6.3](#), we explicitly examine the capacity drawdown on the supply side of the lending market.

Figure 6 plots the lending market utilization rate, defined as the ratio of actual share lending to structural lending capacity ($\ell_{t+1}^O / \kappa s_{t+1}^O$), across the net state space a_t^P .

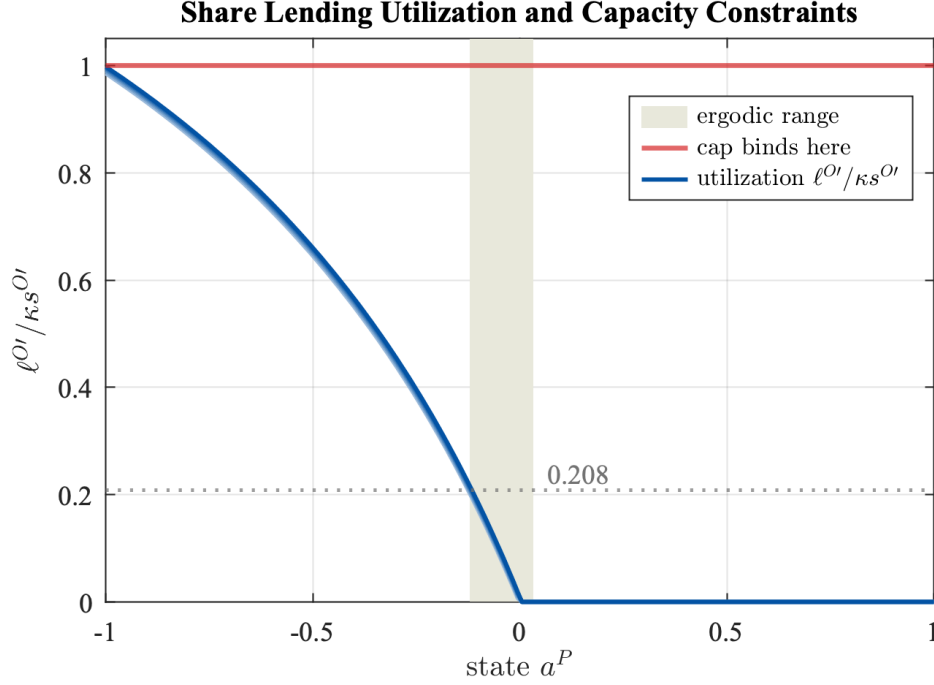


Figure 6: Share Lending Utilization and Capacity Constraints. The lending market utilization rate ($\ell_{t+1}^O / \kappa s_{t+1}^O$) plotted across the net state space a_t^P . Utilization rises with the size of the short and approaches its ceiling only at the far left of the grid; within the shaded ergodic band it peaks at 0.208. Source: `gmod/fee_cost/plot_lending_utilisation.m`.

Utilization, Ergodic Range, Capacity Ceiling. The horizontal axis maps how differing asset allocations consume available market liquidity. For $a_t^P \geq 0$, the lending utilization rate is flat at zero, confirming that no short-selling occurs when the Pessimist retains a positive net asset position. Within the shaded ergodic range ($a_t^P \in [-0.120, 0.033]$), utilization remains comfortably low, peaking at 0.208 on the grid at the band's left edge and falling back toward zero as the state approaches the zero net boundary. This shows that during normal economic states, the Optimist's capacity buffer is highly effective, allowing the market to absorb modest shorting demand with minimal structural friction (lending capacity to spare). However, reading leftward past the ergodic band, the curve behaves as the mechanism implies: successive high dividend draws enrich the Optimist, widen the gap between the market price

and the Pessimist's own valuation, and deepen her short, so utilization climbs steeply and near-concavely toward the ceiling of 1.0 marked by the red boundary.

What Actually Stops the Short. The ceiling in Figure 6 is a property of the state space, and reading it as the thing that halts the short would misplace the mechanism. What halts it is the marginal condition (47): the short runs to the size at which carrying one more unit costs exactly what that unit's mispricing is worth.

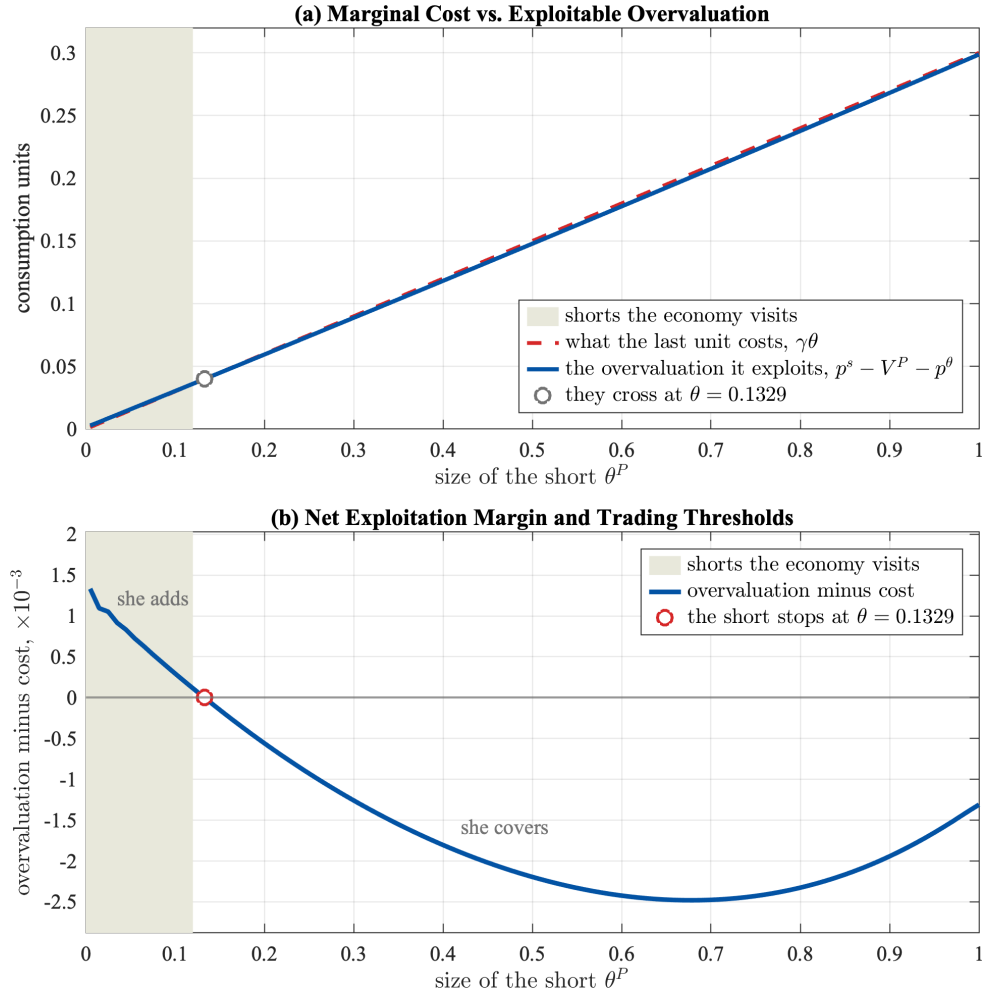


Figure 7: The Shorting Marginal Condition: Equilibrium Position Limits Under Endogenous Frictional Costs. The two sides of (47), evaluated at the highest dividend state and plotted against the size of the position. *Panel (a)*: the cost of the last unit shorted, $\gamma\theta_{t+1}^P$, against the overvaluation it exploits, $p_t^s - V_t^P - p_t^\theta$. *Panel (b)*: that difference. It is positive for small positions, so the Pessimist adds; it crosses zero at $\theta_{t+1}^P = 0.133$; and it is negative beyond, so she covers. Source: `gmod/fee_cost/plot_short_foc.m`.

Figure 7 draws the two sides of (47) against the size of the position, at the highest dividend state so that the pressure to short is as strong as the model allows. They cross at $\theta_{t+1}^P = 0.133$, where the short the condition implies equals the short already held. The overvaluation curve is convex against a cost line of slope γ , and their difference, $(p_t^s - V_t^P - p_t^\theta) - (\gamma\theta_{t+1}^P)$, passes across zero at that same short position.¹¹

Asymptotic Convergence and Slacks. The only term in (47) that grows with the position is $\gamma\theta_{t+1}^P$, while the term of valuation gap does not scale with the short, and the base rate is zero throughout this equilibrium. That is what pins the short at a finite size, so the capacity constraint is reached only if the dividend process can push the state past it. It cannot. Fixing the dividend state makes the transition deterministic, since the solved rule $g(a_t^P, z_t)$ collapses to a one-dimensional map $a_{t+1}^P = g(a_t^P, \bar{d})$ that is simply iterated; forcing the highest state in every period is therefore the most extreme dividend path the model admits, and whatever it reaches bounds what any path can reach. Figure 8 runs that experiment and reveals that lending utilization asymptotically converges to a strict upper bound of just 23.5%, leaving the capacity constraint over 76.5% slack. However, given the ergodic band, the maximum utilization reached anywhere across the full stochastic simulation is bounded at 21.4%.

The carrying cost is therefore what prevents the lending capacity from ever being exhausted, and the two frictions are not additive: on the equilibrium path the cost makes the cap redundant. The constraint still does work, by bounding the state space and thereby the feasible set of Definition 2, but it does so as a boundary the equilibrium respects rather than one it reaches.

¹¹ V_t^P is rebuilt from the solved policies, by evaluating the stored consumption and price rules at next period's state $a_{t+1}^P = g(a_t^P, z_t)$ and taking the expectation under the Pessimist's own transition matrix.

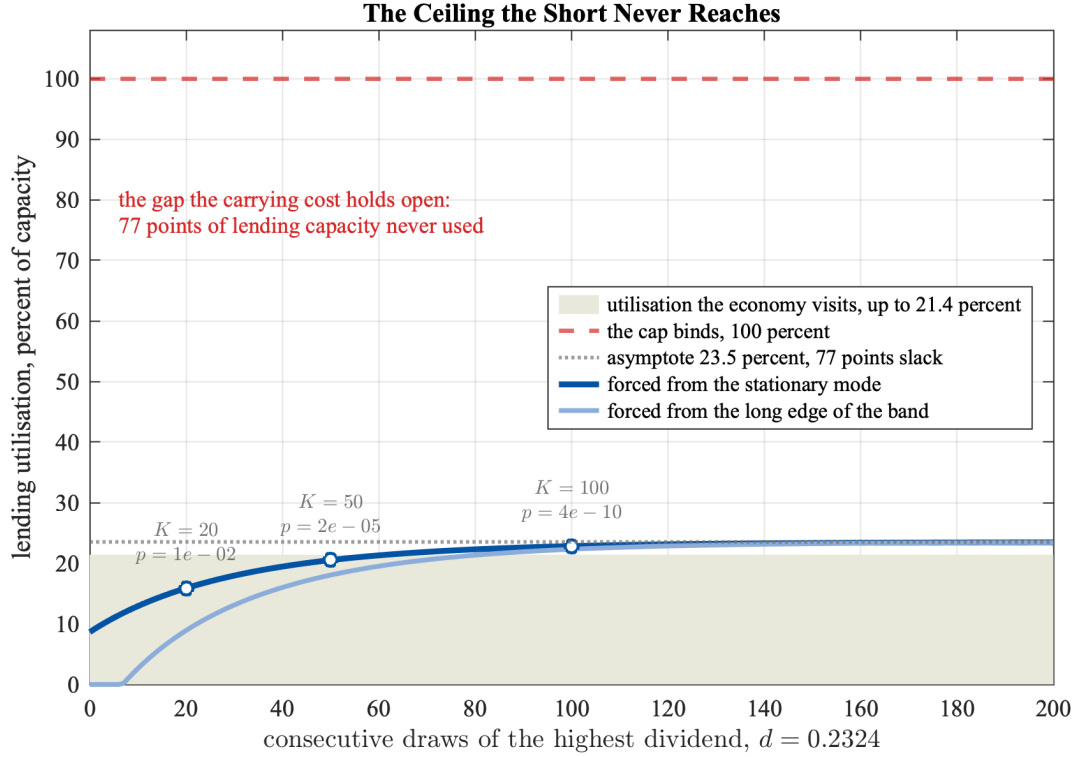


Figure 8: The Ceiling the Short Never Reaches. Lending utilization along a deterministic path that forces the highest dividend state in every period, started at the stationary mode and at the long edge of the ergodic band. Both converge to 23.5% of capacity, leaving the constraint 76.5 points slack, against the shaded region of utilization the simulated economy actually visits. Marked points give the probability of a run of that length, 0.805^K . Source: `gmod/fee_cost/plot_capacity_ceiling.m`, with the same numbers in table form from `gmod/fee_cost/test_high_dividend_run.m`.

6.5 Stability and the Restoring Force

The short does not run away, and nothing in the lending market stops it: the fee is zero and the cap never binds. Its dynamics are governed instead by a restoring force internal to the Pessimist's optimal choice, measured here at one dividend state in Figure 9 and averaged over the dividend chain across the whole grid in Figure 10.

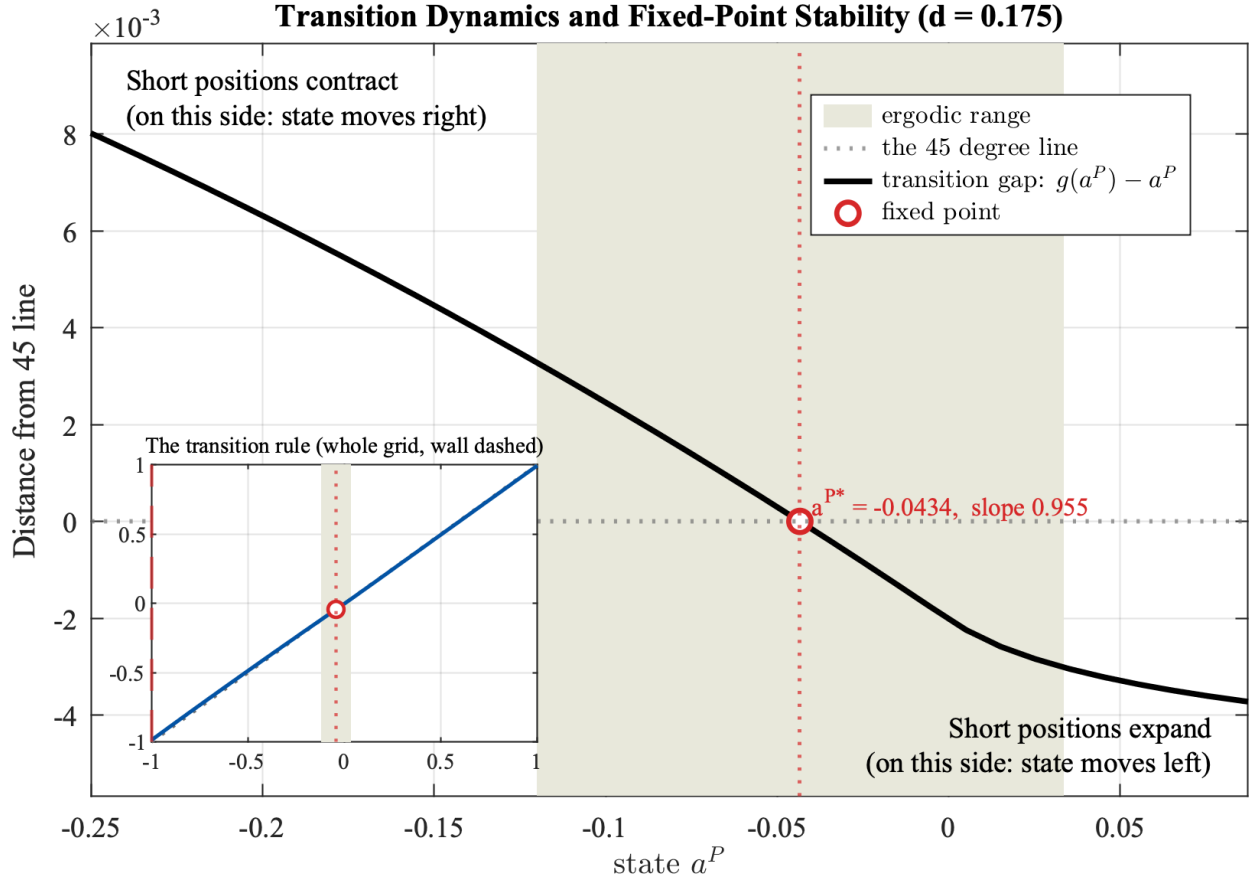


Figure 9: Transition Dynamics and Fixed-Point Stability ($d = 0.175$). The transition gap $g(a_t^P) - a_t^P$ at the median dividend state, drawn so the 45-degree line is the horizontal axis and the fixed point is the crossing. It is one of the five conditional drifts of Figure 10, shown alone. *Inset:* the same rule in levels across the whole grid, feasibility wall dashed. Source: gmod/fee_cost/plot_transition_single.m.

The Conditional Fixed Point. At this median dividend state ($d = 0.1750$), the transition rule features a unique conditional fixed point, denoted by $a^{P*} = -0.0434$, which represents the isolated steady-state attractor where the policy function intersects the 45-degree line

$(g(a^{P*}) = a^{P*})$. This conditional steady state relates directly to the shorting marginal condition in (47). Because the Pessimist holds no long inventory in this negative regime ($s_{t+1}^P = 0$), her portfolio maps directly to a physical short position of $\theta_{t+1}^P = -a^{P*} = 0.0434$. At this exact allocation, the marginal carrying cost $\gamma\theta_{t+1}^P$ precisely balances the subjective overvaluation wedge $(p_t^s - V_t^P) - p_t^\theta$, removing any incentive to adjust her portfolio and setting the active transition drift to zero.

As illustrated in Figure 9, this fixed point operates as a local attractor from both sides of the support. To its left, the transition gap is positive ($g(a_t^P) - a_t^P > 0$): the size premium rises with the position, the marginal cost of carrying the deep short position exceeds the exploitable overvaluation wedge, prompting the Pessimist to cover her excess exposure and forcing the state variable to mean-revert to the right. To its right, the transition gap turns negative ($g(a_t^P) - a_t^P < 0$). In this region, the marginal size premium is suppressed relative to the overvaluation gap, yielding a positive net marginal benefit that induces the Pessimist to expand her short position and drives the next-period state variable to the left. The local policy slope at the intersection is 0.955, satisfying the contraction mapping condition ($|g'| < 1$), which ensures global stability and governs the speed of state mean-reversion. The unconditional ergodic band tightly encompasses this attractor, while the inset confirms that the global policy mapping remains strictly invariant within the physical feasibility walls across the entire support grid.

The Expected Restoring Force and Global Existence. Economically, this restoring force represents the net marginal return on shorting: the gap between the marginal carrying cost of the final unit shorted, $\gamma\theta_{t+1}^P$, and the subjective overvaluation wedge it exploits, $(p_t^s - V_t^P) - p_t^\theta$, as formalized by the shorting marginal condition in (47). The Pessimist covers her short exposure when marginal costs exceed this wedge, adds to her position when costs fall short, and rests where they equate. This stabilizing force guarantees global equilibrium existence; in a frictionless economy ($\gamma = 0$), this stabilizing margin vanishes, triggering a

severe boundary divergence where the state variable breaches the feasibility bounds and destroys the invariance of any compact state support (Appendix F).

Numerically, Figure 10 plots the unconditional one-period expected drift, $\mathbb{E}[a_{t+1}^P - a_t^P]$, integrated across the objective dividend transition matrix Π^P . The unconditional resting point, denoted as the endogenous coordinate a^{P° satisfying

$$\mathbb{E}[a_{t+1}^P \mid a_t^P = a^{P^\circ}] = \sum_{k=1}^N \pi_k^P g(a^{P^\circ}, d_k) = a^{P^\circ}$$

is located at $a^{P^\circ} = -0.0408$, where π_k^P represents the stationary probability of dividend state d_k . At this coordinate, the expected marginal carrying cost for next period exactly balances the expected subjective overvaluation wedge today, reducing the net structural drift to zero.

However, the non-linear trajectory reveals that the expected restoring force is highly non-monotonic in short position intensity. The expected rightward pull is 0.0033 at the lower boundary of the ergodic core, accelerates non-linearly to a peak of 0.0173 near $a_t^P = -0.79$, and subsequently attenuates as the state variable approaches the absolute physical boundary wall. The structural friction thus operates as an asymmetric, hump-shaped policy brake rather than an explosive boundary asymptote. This maps directly to the quantity-side implications derived in Section 6.4: the endogenous restoring force exerts its maximum stabilization intensity well before the physical inventory ceiling is reached, preventing any equilibrium dividend path from driving the economy to the hard constraint.

Synthesis. This section characterizes the structural lending mechanism: subjective disagreement generates short-selling volume; this short interest draws down lendable inventory; because carrying costs are strictly convex, position expansion escalates the per-share size premium; and this rising premium acts as an endogenous brake that restricts optimal short allocations to a bounded, stable ergodic set. Having mapped these static policy surfaces, Section 7 sets the system in motion.

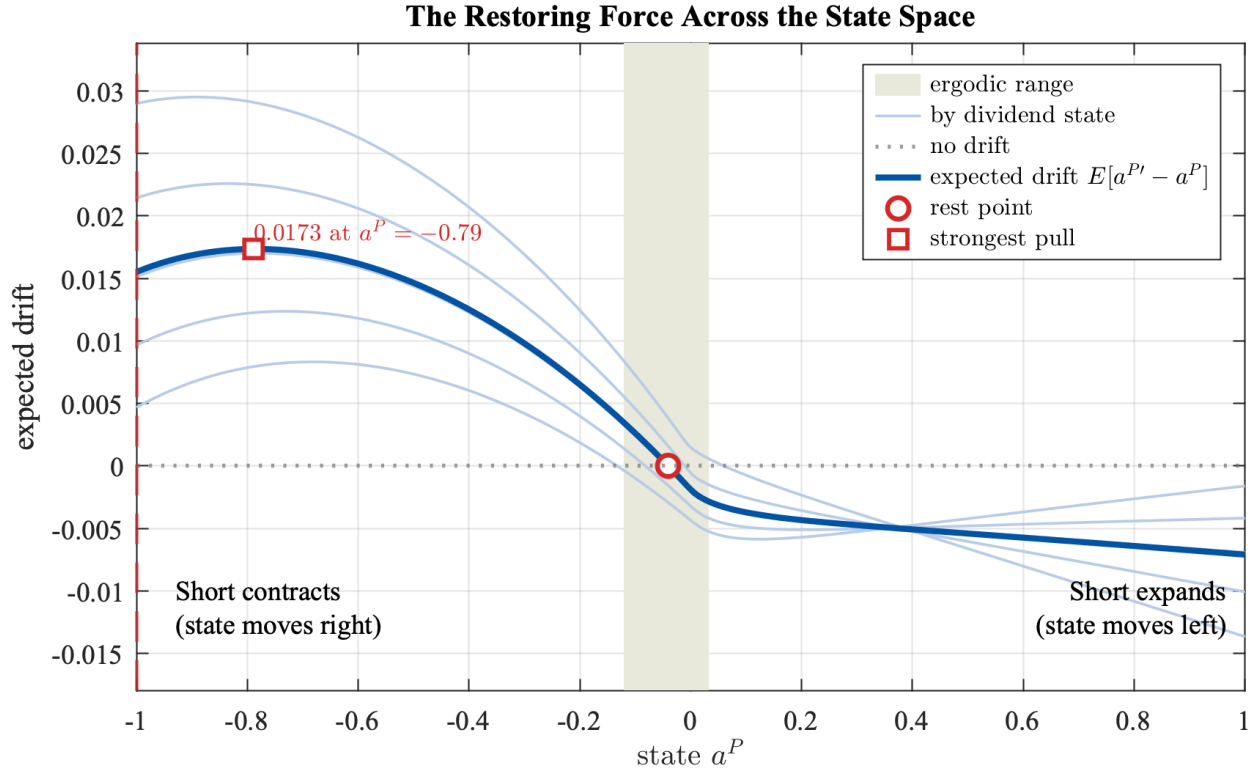


Figure 10: The Expected Restoring Force Across the State Space. The unconditional one-period expected drift $\mathbb{E}[a_{t+1}^P - a_t^P]$ (solid blue line), representing the globally solved transition rule integrated over the objective dividend chain. The five underlying conditional drift schedules are plotted in pale blue, corresponding to each discrete dividend state. Above the zero reference line (dotted), the short position contracts (the state moves rightward); below it, the short position expands (the state moves leftward), with the zero-crossing identifying the unconditional rest point $a^{P^o} = -0.0408$. The conditional attractors span from $+0.0593$ at the minimum dividend node to -0.1331 at the maximum node, with Figure 9 isolating the median state transition schedule. Source: `gmod/fee_cost/plot_phase_drift.m`.

7 Equilibrium Dynamics and Counterfactuals

While [Section 6](#) characterized the equilibrium via state-space mappings that detail policies for asset prices, leverage, and expected drift, this section evaluates how those structural rules manifest over time. We first trace the model’s dynamic transitions by imposing a forced dividend run ([Section 7.1](#)), then analyze endogenous fluctuations by simulating unconstrained dividend paths ([Section 7.2](#)). Finally, [Section 7.3](#) executes counterfactual experiments by varying the carrying-cost intensity parameter (γ). This analysis evaluates how the model’s global stability boundaries and asset-pricing predictions respond when the internal restoring force is varied, establishing a falsifiable test of the model’s core mechanism.

7.1 The Delayed Dynamics of Shorting Costs

Because the endogenous state variable adjusts more slowly than the dividend process that drives it, an isolated draw and a sustained run generate fundamentally distinct transitional dynamics, and only the latter can reproduce the multi-period episodes documented in [Section 3](#).

Shock Persistence and the Separation of Timescales. To capture the structural mechanics documented in the empirical event study, the model requires persistent cash flow runs rather than isolated transitory innovations. A single, isolated dividend shock achieves negligible amplification. This minor impact stems from an internal mismatch in velocity between the exogenous and endogenous processes: the fundamental dividend deviation exhibits a half-life of 8 periods, whereas the sluggish endogenous state variable requires approximately 16 periods to close half of its structural gap.¹² Consequently, a brief shock

¹²Both half-lives measure the geometric time required to halve a given structural deviation. For the dividend process, the chain is initiated at its maximum support node ($d = 0.2324$) and iterated toward its stationary mean ($\bar{d} = 0.1750$). The initial deviation of 0.0574 is halved when $\mathbb{E}[d_t]$ crosses 0.2037, which occurs at $t = 8$. For the state variable, the averaged transition mapping defined in [Section 6.5](#) is iterated forward from a trough of -0.0916 ; the halfway marker to the unconditional rest point (-0.0408) is -0.0662 , which the system first intersects at $t = 16$. Because both paths are geometric, neither gap closes entirely; the

forces the state variable to chase a moving target that mean-reverts faster than the portfolio can rebalance, preventing any meaningful accumulation of balance-sheet stress or shorting premium spikes.

Replicating the empirical data requires a sustained macroeconomic event. We therefore generate a run by holding the dividend at its top state for twenty-two periods, long enough for the state to accumulate and short enough to stay well inside the ergodic set. That gives the state the runway it needs, and [Figure 11](#) traces what follows. The position troughs at -0.0916 at $t = 26$, four periods after the run ends, because it is still closing the gap to a target that has already turned. The size premium rises from 0.356% to 0.696% , a factor of 1.96 , so the run reproduces the shape of the empirical ramp without matching its magnitude.

Price Decomposition and the Short's Own Contribution. Panel (b) separates the price into the part the dividend would have moved and the part the short is responsible for. Writing $p^s(a_t^P, z_t)$ for the equilibrium price as a function of the position and the dividend state, so that $p_t^s = p^s(a_t^P, z_t)$ along the realized path, and holding the position at the stationary mean -0.0424 from which the path starts, we decompose the total price deviation as follows:

$$p^s(a_t^P, z_t) - p^s(a_{\text{mean}}^P, z_0) = \underbrace{[p^s(a_{\text{mean}}^P, z_t) - p^s(a_{\text{mean}}^P, z_0)]}_{\text{the dividend, in grey}} + \underbrace{[p^s(a_t^P, z_t) - p^s(a_{\text{mean}}^P, z_t)]}_{\text{the short, shaded}}.$$

The two channels act at disjoint times. At $t = 0$, the position is predetermined and has not moved, so the price's jump from 3.580 to 4.136 is driven by the dividend alone. From $t = 1$ to $t = 22$, the dividend is held constant and cannot move again, meaning the entirety of the remaining upward clamber is manufactured by the short position: the short's contribution reaches $+0.0599$ by the end of the run, representing 9.7% of the total baseline-to-peak price run-up.¹³

half-life provides an exact invariant summary, as tabulated across alternative calibrations in [Table 5](#).

¹³The dividend leg is the jump at $t = 0$, $4.1355 - 3.5803 = 0.5552$; the short leg is everything after it, $4.1954 - 4.1355 = 0.0599$, which is also the gap between the two curves at $t = 22$, the counterfactual being flat while the dividend is held. The two add to 0.6150 , so $0.0599/0.6150 = 9.7\%$.

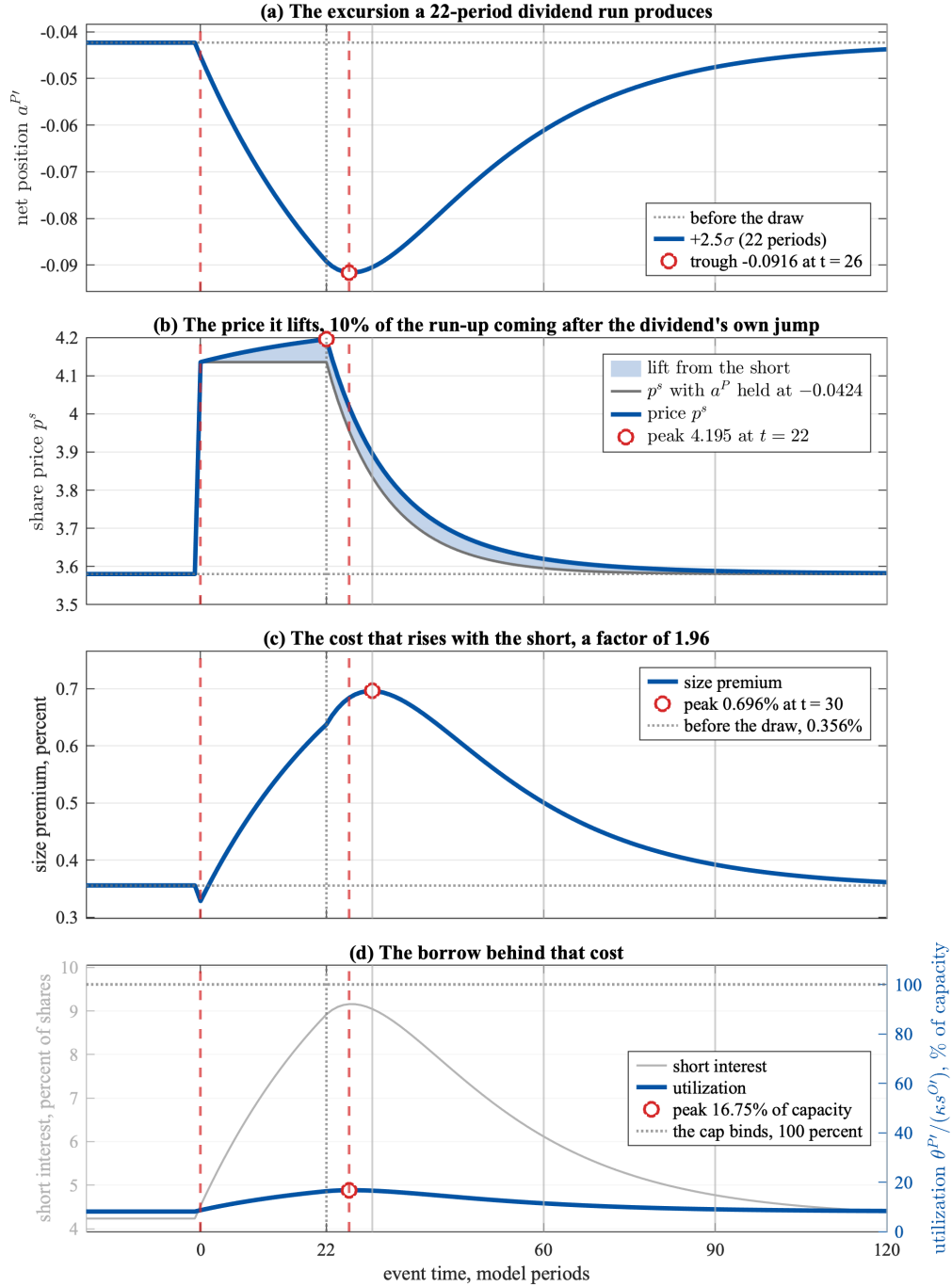


Figure 11: Economic Responses to a 22-Period Maximum Dividend Run. Paths show exact conditional expectations tracking the economy from its stationary mean under a sustained $+2.5\sigma$ shock sequence. Panels track: (a) the deep excursion of the net position a_{t+1}^P to a trough of -0.0916 , (b) the share price, where pinning the position at that same stationary mean (-0.0424) isolates the dividend channel, revealing that the short's own contribution drives 10% of the run-up to a peak of 4.195, (c) the resulting 1.96-fold surge in the size premium, and (d) short interest and capacity utilization expansions. Panels (a), (c) and (d) all read the position the Pessimist chooses at each state, $\theta_{t+1}^P = \text{neg}(a_{t+1}^P)$, which is the quantity the premium and utilization are defined on. Source: `gmod/fee.cost/plot_dividend_impulse.m`.

Why the Short’s Contribution Is Positive. The sign of that contribution is the core mechanism rather than an asset-pricing anomaly, and two structural identities dictate it. Because the Optimist is the marginal holder, $p_t^s = V_t^O$ holds throughout the trajectory. Every share borrowed and sold is a synthetic share delivered to the agent who values the asset the most:

$$s_{t+1}^O = 1 + \theta_{t+1}^P - s_{t+1}^P$$

This quantity exceeds the standard physical float, implying that a deeper short position forces the Optimist to absorb more equity. Read from the Pessimist’s side, (47), the equity valuation with a base rate of zero yields:

$$p_t^s = V_t^P + \gamma \theta_{t+1}^P$$

The price stands above her own valuation by exactly the marginal cost of carrying the last unit she shorted. She continues to sell short until the carrying cost equals the overvaluation the unit exploits and then stops; thus the mispricing that survives in equilibrium is the friction itself. Short-selling in this framework therefore relocates the float toward the optimist and leaves a wedge rather than closing one. The relation is explicitly local rather than global, and the qualification is one about the position rather than about time: read as a function of the position, [Section 6.3](#) establishes that the price schedule $p^s(a^P)$ peaks and turns down once the quadratic charge overtakes the linear benefit. That peak lies well outside the ergodic band, so within the region the economy visits a deeper short always raises the price. The downturn traced in [Figure 11](#) is a different object: it is the path over time, and it turns when the dividend run ends rather than when the quadratic charge binds.

The Ordering of the Three Peaks. The paths in panels (a)–(c) reveal two distinct temporal mismatches that confirm the endogenous short-market friction possesses its own structural momentum, decoupling asset prices from contemporaneous news. The equity price

peaks at $t = 22$, when the dividend run terminates. The position goes on deepening to $t = 26$, and the premium keeps climbing to $t = 30$: it is $\gamma\theta_{t+1}^P/p_t^s$, so a position still growing and a price already falling both push it up, and it turns only once the recovering position outweighs the still-falling price. The cost of carrying the short therefore keeps rising for eight periods after the price has peaked and begun to fall. This is a phenomenon driven by the fact that short positions continue to expand sluggishly, driving lending utilization further up even as equity prices decline, decoupling the peak of the carrying cost from the peak of the equity price. Utilization peaks at a modest 16.75% (Panel d), so the hard capacity cap never binds and the base rate fee remains flat at zero. What moves is the other charge: those extra shares are pushing further up a convex cost function, and that is what carries the size premium to its maximum of 0.696% on a delayed timeline. The other asymmetry is visible at the opening of the episode, $t = 0$, where it produces a feature of panel (c): the premium, $\gamma\theta_{t+1}^P/p_t^s$, falls on impact before it begins to climb. That is because a high dividend draw lifts that denominator long before the position it provokes can catch up.

The lending market therefore stays expensive long after the news that made it expensive has passed: borrowing costs are high not because anything new is happening but because the position that raised them has not yet been unwound. That a lending market adjusts slowly is not itself new; what matters here is that the delay is generated by the equilibrium rather than imposed on it, the distinction that literature draws against the search and slow-moving-capital accounts (Section 2.3).

7.2 Endogenous Event-Time Dynamics

The 22-period dividend run above is a deliberate experiment: it forces the highest dividend state in every period. We next ask whether comparable episodes arise when the economy is left to evolve under its own stochastic process. A 100-path, 10,000-period simulation under Π^P , and dating one episode per path by its own fee peak as Section 3 does in the data, generates 98 endogenous high-dividend episodes that match the empirical fee-peak window

structure.

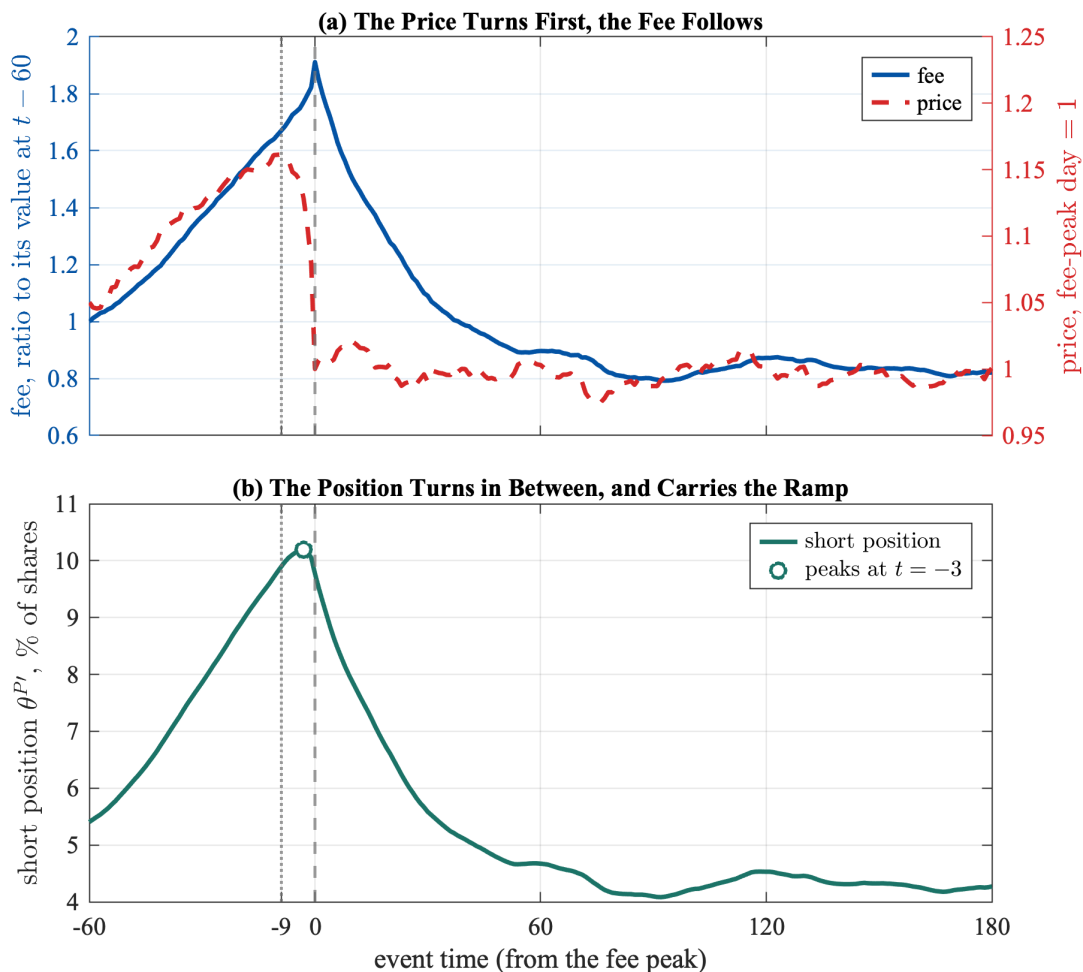


Figure 12: The Model's Average Episode. Both panels use the same 98 simulated episodes, aligned by the fee peak. Panel (a) plots the fee relative to $t = -60$ and the price relative to its value at the fee peak. The price is normalized to one at $t = 0$; its maximum is 1.16 at $t = -9$. Panel (b) shows the short position, which peaks at $t = -3$. Thus, the three turning points occur in the order price, position, fee, as in Figure 11, but emerge endogenously here. The dates -9 and -3 characterize the averaged path. Source: `gmod/fee_cost/plot_model_episode.m`.

Three Features of the Empirical Pattern. Figure 12 shows three features of the empirical event-time pattern that emerge endogenously by the simulation. The fee builds into its peak over many periods rather than jumping to it, and falls back afterwards. The price stands above its fee-peak-day level throughout the run-up, so it falls into the fee peak rather than rising into it. And the two do not turn together: the price peaks at $t = -9$ while the fee goes on climbing to $t = 0$. The dates -9 and -3 characterize the averaged path.

Measured episode by episode, the median lag between the price turn and fee peak is four periods, and the price turns first in 97 of the 98 episodes.

The Same Ordering, Unforced. Panel (b) identifies what fills the gap, and it is the same mechanism isolated by the forced run. The short position peaks at $t = -3$, after the price turns and before the fee peaks. Thus, the three turning points occur in the order: price $\rightarrow$ short position $\rightarrow$ fee, which is the central dynamic implication of the mechanism. This ordering is not a coincidence of timing: it reflects the separation of timescales isolated by the forced run, now emerging endogenously with the dividend process left unconstrained. [Section 3](#) documents the same ordering in the data over a longer horizon: the price is already near its maximum when the 60-day window begins.

Model vs. Data Moments. [Table 4](#) evaluates the model out-of-sample; parameters of [Table 2](#) are set entirely by preferences rather than fitted to the event study. Although the model correctly replicates the qualitative dynamics (specifically the price decline into the fee peak, a positive fee ramp, subsequent decay, and negative returns over the event window), it exhibits quantitative shortfalls. The simulated fee ramp captures only 55% of its empirical counterpart, while the model generates 66% of the absolute raw price drop, just 32% of the underperformance net of the S&P 500, and exhibits a slower fee decay.

Moment	Data	Model	
Fee ramp, peak / baseline	3.481	1.911	55% of the data
Fee decay, $t+120$ / peak	0.212	0.457	decays too slowly
Return, $t-60$ to $t+120$, raw	-7.2%	-4.7%	66% of the fall
Return, $t-60$ to $t+120$, net of S&P	-14.9%	-4.7%	32% of the fall
Price direction into the fee-peak	falls	falls	correct sign

Table 4: The Model Beside the Event Study. The data column is from [Table B.1](#); the model column is computed from the 98 simulated episodes in [Figure 12](#). Threshold sensitivity details are provided in [Appendix B.3](#). Source: `gmod/fee_cost/export_model_tables.m`, from `data/model/solve_fee_cost.mat` and `data/stylized_facts/empirical_composite.csv`.

What Drives the Fee Ramp. The panel also identifies which component drives the fee ramp in these simulated episodes. Rebuilt from the two averaged series as $\gamma \bar{\theta}_{t+1}^P / \bar{p}_t^s$, the charge decomposes exactly into a position component and a price component:

$$\underbrace{+0.6398}_{\Delta \log \text{ charge}} = \underbrace{+0.5927}_{\Delta \log \theta_{t+1}^P} - \underbrace{(-0.0471)}_{\Delta \log p_t^s}.$$

From $t = -60$ to the fee-peak, the position accounts for 93% of the increase, with the price accounting for the remaining 7%.¹⁴ The short position rises from 5.4% to its own peak at 10.2% of shares, a factor of 1.89, closely matching the 1.91-fold increase in the fee. Thus, the tightening in borrowing costs is driven primarily by the growth of the short position.

7.3 Counterfactual Analysis: Varying the Carrying Cost

While the carrying cost stabilizes the short position along the equilibrium path, time dynamics isolate its role poorly because the premium, short interest, and position move together endogenously. To test this mechanism, we depart from the equilibrium path by freezing every counterfactual economy at its deepest crisis trough ($a_t^P = -0.0916$, reached during a 22-period dividend run) and re-solving across the permissible parameter range $\gamma \in [0.10, 0.50]$. Within this compact subset, a competitive equilibrium exists; outside it, the policy functions cross the feasibility boundary (Appendix D.1).

Effects on the Structural Rest Point. As shown in Table 5, varying γ shifts the economy's structural rest point. A weak charge ($\gamma = 0.10$) pushes the rest point below the trough, yielding a negative instantaneous drift ($\mathbb{E}[\Delta a_t^P] = -0.00054$) that causes the short position to deepen further. Conversely, a steeper charge ($\gamma \geq 0.20$) lifts the rest point clear

¹⁴The position goes from 5.403% to 9.773% of shares, so $\Delta \log \theta_{t+1}^P = \log(9.773/5.403) = +0.5927$; the price goes from 3.784 to 3.609, so $\Delta \log p_t^s = -0.0471$. By summation, the charge is +0.6398, and the shares are $0.5927/0.6398 = 92.6\%$ and $0.0471/0.6398 = 7.4\%$. The decomposition is exact for the charge rebuilt from the averaged series; the fee drawn in panel (a) averages the charge episode by episode instead, so it ramps by 1.911 against the 1.896 of the series decomposed here, a mean of ratios against a ratio of means.

of the trough, generating a positive drift that accelerates the half-life recovery back toward the baseline state.

Carrying-cost coefficient (γ)	Marginal carrying cost (% of price)		Drift at the trough	Zero-drift rest point	Stationary mean	Half back (periods)
	At the trough	At stationary mean				
0.10	0.2603	0.3397	−0.00054	−0.1160	−0.1209	<i>never</i>
0.20	0.5044	0.3523	+0.00108	−0.0604	−0.0629	20
0.30 (baseline)	0.7394	0.3557	+0.00225	−0.0408	−0.0424	16
0.40	0.9675	0.3561	+0.00318	−0.0308	−0.0319	13
0.50	1.1903	0.3551	+0.00396	−0.0246	−0.0255	12
Elasticity in γ	+0.94	+0.03	—	−0.96	−0.97	—

Table 5: Carrying Cost Counterfactuals at the Crisis Trough. Evaluates alternative economies frozen at the deepest position reached during the baseline 22-period dividend run ($a_t^P = -0.0916$) across varying cost intensities (γ). Columns report the marginal carrying cost $\gamma\theta_{t+1}^P/p_t^s$ at both the trough and the stationary mean, instantaneous drift ($\mathbb{E}[\Delta a_t^P]$), zero-drift rest points, position stationary means, and recovery half-lives. The near-zero elasticity at the stationary mean (+0.03) versus the steep position contraction elasticity (−0.97) demonstrates that the carrying cost functions as a quantity constraint rather than a price driver in general equilibrium. The base rate is zero throughout the solved equilibrium, so the quoted borrowing fee f_t/p_t^s is half these figures: 0.178% at the stationary mean of the calibrated $\gamma = 0.30$, and the same +0.03 elasticity, a constant factor cancelling in a log ratio. At ultra-low cost intensity ($\gamma = 0.10$), the instantaneous drift becomes negative because the rest point shifts below the crisis trough itself. Source: `gmod/fee_cost/plot_gamma_counterfactual.m` outputting to `data/model/gamma_counterfactual.csv`.

Separating Fee Rate and Quantity Dynamics. When economies are released from the common trough to evolve naturally, γ governs the depth of the trough rather than its borrow rate. Across a fivefold increase in γ , the stationary mean of the short position contracts by a factor of 4.75 (a near-unit elasticity of −0.97). Because this quantity contraction almost perfectly offsets the higher cost intensity, the marginal carrying cost ($\gamma\theta^P$) stays nearly constant. Furthermore, while the asset valuation wedge fluctuates dynamically over time in response to cash-flow and belief shocks, its structural boundaries are anchored in general equilibrium by the fundamental disagreement parameter rather than the friction intensity. The equity price level adjusts proportionally across counterfactuals to preserve this structural spread, keeping the stationary size premium ($\gamma\theta^P/p_s$) evaluated at these means virtually

flat (an elasticity of +0.03). This confirms that the carrying cost coefficient is primarily an instrument for short exposure volume: a steeper charge intensity is absorbed by a smaller position rather than a higher fee rate.

Parameter Identification and Trade-offs. Because the long-run borrow rate is nearly invariant to γ , the coefficient cannot be recovered from observed fee levels and must be identified through position dynamics. While the parameter shifts the path trajectory rather than its height, its impact on the fee profile is dwarfed by its impact on quantities: over the same range, the fee ramp has an elasticity of only +0.19 (Table D.3) compared to the position’s structural contraction elasticity of -0.97 (Table 5). This rigidity ensures that the empirical fee shortfalls cannot be resolved by simply tuning the coefficient. Re-solving the event study across γ values reveals that the model targets themselves cannot be reconciled with the data. Although a higher γ pushes both targets toward the empirical benchmarks, the model hits a rigid structural boundary. At the maximum permissible value ($\gamma = 0.50$), the simulated ramp stalls at 2.058 against the empirical 3.481, while the decay remains trapped at 0.437 against the empirical 0.212. Trying to force a closer fit along this boundary severely degrades the model’s out-of-sample asset pricing accuracy, as the raw return’s magnitude contracts with an elasticity of -0.51 , rapidly carrying it away from the empirical target. Consequently, no single parameter choice can structurally reconcile the core validation moments without undermining predictive validation on asset returns.

The Dual Capacity of the Carrying Charge. The restoring force in this framework is driven entirely by the carrying cost rather than the lending cap, which is never bound (Section 6.4). The charge operates in two crucial capacities: it holds the asset price above the Pessimist’s valuation during the run-up, and it provides the necessary mean reversion across the state space. Stripping the model of both charges yields clean numerical convergence but produces policies that violate feasibility limits (Appendix F). The friction that opens the valuation wedge is ultimately the same friction that bounds the economy.

Synthesis: Structural Implications of the Shorting Fee. Ultimately, varying the friction intensity reveals that the shorting fee operates under a strict general equilibrium insulation property. Rather than functioning as a direct price elevator, a higher shorting fee coefficient acts primarily as a quantity brake: short sellers aggressively contract their exposure volume to offset the steeper structural penalty, leaving the steady-state borrowing fee rate insulated. Conversely, a lower fee coefficient expands short exposure volumes and dangerously weakens the system's mean-reverting dynamics. This structural asymmetric response establishes that the carrying cost is working in a dual capacity. The exact same friction responsible for allowing a speculative valuation wedge to open between Optimists and Pessimists is simultaneously the vital stabilizing anchor that prevents short positions from exploding past feasibility limits, causing the model's mathematical equilibrium to collapse.

8 Conclusion and Implications

This paper studies how short-selling frictions interact with investor beliefs to shape short exposure, borrowing costs, and asset prices. We combine an event study of equity fee-spike episodes with a dynamic heterogeneous-belief general equilibrium model featuring short selling and securities lending. The empirical evidence shows a systematic dynamic pattern: borrowing fees build up over time while stock prices are already declining, and after fees reach their peak and begin to fall, the affected stocks continue to underperform before their prices eventually stabilize. The model provides a structural interpretation of this pattern through the endogenous adjustment of short exposure, with the carrying cost generating both a gradual adjustment in short positions and an eventual stabilizing force for asset prices.

The analysis yields three main results. First, a convex, position-dependent carrying cost generates an endogenous restoring force for short exposure. As short exposure increases, its marginal carrying cost rises, limiting further expansion and inducing the position to contract when the valuation opportunity narrows. Because physical lending capacity remains slack along the equilibrium paths studied, this restoring force operates through the position-dependent carrying cost rather than a binding quantity constraint. The same friction that limits arbitrage therefore also contributes to the mean reversion of short exposure. Through its effect on equilibrium short positions, this restoring force generates an eventual stabilization of asset prices following the decline documented in the empirical episodes.

Second, the carrying-cost intensity parameter primarily affects the quantity of short exposure rather than the equilibrium borrowing rate. Increasing the carrying-cost parameter substantially reduces the equilibrium amount of short exposure while having only a small effect on the borrowing rate. This result reflects a general-equilibrium adjustment: changes in the marginal cost of shorting are absorbed primarily through changes in the quantity of short exposure rather than through proportional changes in its market price.

Third, the model provides a mechanism for the delayed response of borrowing fees to

movements in asset prices. Because short exposure adjusts gradually, it acts as a slow-moving state variable through which persistent fundamental movements accumulate into larger positions. When fundamentals reverse, the asset price responds more quickly than the accumulated short position. In the model's endogenous event-time simulations, the turning point in the asset price precedes that in short exposure, which in turn precedes the peak in the borrowing fee. This ordering provides a structural mechanism for the empirical finding that borrowing fees can remain elevated and continue rising after the stock price has already turned downward.

The analysis also identifies an important quantitative limitation. The smooth, position-dependent carrying-cost specification captures the qualitative timing of the empirical fee spikes but does not reproduce their full magnitude. In particular, increasing the carrying-cost intensity parameter changes short exposure substantially while producing only limited variation in the borrowing rate and fee ramp. The results therefore suggest that additional features of securities-lending markets may be needed to account for the extreme borrowing-cost episodes. Possible extensions include discrete changes in lending capacity, lender recalls, or institutional participation. For the model presented here, a natural modification would be to make the carrying-cost intensity parameter endogenous to market tightness, thereby allowing search and matching frictions to amplify borrowing fees independently of the absolute scale of the aggregate short position.

These findings also have implications for risk management. Borrowing fees should not be interpreted in isolation as a measure of the cost or intensity of short-selling pressure. Because the carrying-cost parameter primarily operates through the quantity of short exposure, changes in borrowing conditions can be accompanied by substantial adjustments in positions rather than proportionate changes in the borrowing rate. Moreover, the delayed adjustment of short exposure implies that borrowing fees can remain elevated after asset prices have already turned. Monitoring borrowing costs together with the evolution of short exposure can therefore provide a more informative view of the persistence of short-selling pressure

than either measure alone. The quantitative limitation of the model further suggests that extreme borrowing-cost episodes may reflect institutional features of securities lending that are not captured by a smooth, position-dependent carrying cost.

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Appendix

A Terminology and Notation

Concept	Term
The capacity component	
$p_t^\theta \theta_{t+1}^P$	basic fee
p_t^θ	base rate
The carrying-cost component	
$\frac{1}{2}\gamma(\theta_{t+1}^P)^2$	carrying cost
$\frac{1}{2}\gamma\theta_{t+1}^P$	size premium
$\gamma\theta_{t+1}^P$	marginal carrying cost
Rates, states, and the wedge	
$f_t = p_t^\theta + \frac{1}{2}\gamma\theta_{t+1}^P$	total quoted borrow rate
f_t/p_t^s	shorting / borrowing / lending fee (a rate on price)
$\frac{1}{2}\gamma\theta_{t+1}^P/p_t^s$	size premium (a rate on price)
$\gamma\theta_{t+1}^P/p_t^s$	marginal carrying cost (a rate on price)
γ	carrying-cost intensity
θ_{t+1}^P	short exposure
$p_t^s - V_t^P - p_t^\theta$	net valuation wedge

Table A.1: Terminology and notation.

B Data

B.1 Episode Construction and Sample Attrition

Of the 3,834 tickers in the original panel, 1,526 ever cross the 10% APR threshold. Applying the 60-day pre-window filter leaves 1,185 tickers, and enforcing the 120 to 180-day post-window filter further reduces the sample to 406 episodes. The primary cause of sample attrition is the post-window constraint. Because the panel ends on 6 December 2023, any fee peak occurring during the final 180 trading days lacks the necessary forward data and is dropped.

From the remaining 406 events, one final ticker (UNAM) is manually excluded to arrive at the 405 episodes used in the paper. Although UNAM’s fee peaked at 15.7% APR on 28 February 2023, its subsequent suspension was driven by a regulatory intervention when California’s Insurance Commissioner placed its sole operating subsidiary into conservatorship. Furthermore, its final three trading quotes on Nasdaq contained severe data errors, including an erroneous 161-fold price jump (from \$0.85 to \$137) right before it was reassigned as a shell company and resumed over-the-counter trading at \$0.12. This episode represents a regulatory seizure and trading halt rather than a market-driven shorting spike, justifying its exclusion.

B.2 Price Indices and Composite Paths

Index the 405 episodes by i and event time by τ , running from -60 to $+180$ trading days with $\tau = 0$ the day episode i ’s own fee peaks. Let P_τ^i be the closing price of the security in episode i at event time τ , and X_τ^i the S&P 500 level on the calendar date that event time corresponds to for that episode. Because episodes peak on different calendar dates, the benchmark at a given τ differs across i .

Write A_τ^i for the *absolute* price index of episode i , which tracks the security’s own price, and R_τ^i for its *benchmark-relative* counterpart, which tracks that price measured against the S&P 500. Both are indices rather than returns: each divides through by the episode’s own

peak-day value, so every episode enters at one when $\tau = 0$ and a reading of 0.90 means ten percent below the peak-day level.

$$A_\tau^i = \frac{P_\tau^i}{P_0^i}, \quad R_\tau^i = \frac{P_\tau^i/X_\tau^i}{P_0^i/X_0^i}. \quad (\text{B.1})$$

Averaging across episodes at each event time gives the two *composite paths*, written $\bar{A}_\tau$ and $\bar{R}_\tau$, that [Figure 1](#) plots. They are equal-weighted, and their bands are standard errors from cross-episode dispersion:

$$\bar{A}_\tau = \frac{1}{N} \sum_{i=1}^N A_\tau^i, \quad \text{s.e.}(\bar{A}_\tau) = \frac{1}{\sqrt{N}} \text{sd}_i(A_\tau^i), \quad N = 405, \quad (\text{B.2})$$

with $\bar{R}_\tau$ defined the same way from R_τ^i . Every episode counts once regardless of the size of the firm or the length of its price history beyond the window. No smoothing is applied: each plotted point is a cross-section over the 405 episodes at that event day, and the smoothness of the curves comes from averaging rather than from a filter.

Rearranging R_τ^i shows what the benchmark-relative measure is:

$$R_\tau^i = \frac{P_\tau^i/P_0^i}{X_\tau^i/X_0^i} = \frac{A_\tau^i}{X_\tau^i/X_0^i}. \quad (\text{B.3})$$

It is the security's price path *divided by* the benchmark's, each normalized to its own peak-day value, and not the difference between two returns. For $\tau > 0$ both ratios are gross returns from the peak day forward, so R_τ^i is then the ratio of the security's gross return to the benchmark's; for $\tau < 0$ neither is a return, since the base date lies ahead of the observation rather than behind it. The distinction is worth stating because the two are not equal and the paper quotes the ratio. At $\tau = +180$ the composite gives $\bar{A} = 1.0141$ and $\bar{R} = 0.8971$, which imply an average benchmark growth of $\bar{A}/\bar{R} = 1.1304$ over the event window. The ratio measure reports -10.29% , whereas an arithmetic excess return would report $1.41 - 13.04 = -11.63\%$. Both are defensible; the ratio is used throughout, including in [Table B.1](#), because it is the

measure that normalizes to one on the fee-peak day and so is directly comparable with A_τ^i in the same figure.

	Quantity	Value	s.e.
Fee	baseline, $t - 60$	16.51% APR	1.84
	peak, $t = 0$	57.46% APR	5.65
	at $t + 120$	12.18% APR	1.20
	peak / baseline	3.481	
	decay, $t+120$ / peak	0.212	
	half-life	2 days	
Price, net of S&P	at $t - 60$ (peak = 1)	1.0773	0.0268
	at $t + 180$	0.8971	0.0236
	post fee-peak day return	-10.29%	
	return, $t - 60$ to $t + 120$	-14.86%	
Price, raw	at $t - 60$ (peak = 1)	1.0790	0.0271
	at $t + 180$	1.0141	0.0269
	post fee-peak day return	+1.41%	
	return, $t - 60$ to $t + 120$	-7.22%	

Table B.1: The empirical moments. Fee and price quantities averaged across the 405 episodes over the 241-day event window, with standard errors where the entry is an average rather than a ratio. The moments the model is scored against are set out in [Table 4](#); the two $t - 60$ to $t + 120$ returns are the rows it uses, while the post fee-peak day returns run to $t + 180$. Source: `code/export_moments_table.m`.

One consequence of averaging ratios rather than taking a ratio of averages is that $\bar{A}_\tau$ is not the growth factor of any single portfolio. It is the mean experience of an episode, which is the object the model is asked to reproduce.

B.3 Sensitivity to the Fee Cutoff

[Table B.2](#) evaluates how the event sample responds to the discretionary choice of the fee cutoff threshold. Across rows running from no cutoff to a peak above 50% APR, the average peak fee increases more than sixfold, from 23.8% to 154.6% APR. Despite this substantial variation in levels, the two composite fee moments, the ramp ratio and the decay, remain remarkably stable. The ramp ratio varies narrowly between 3.35 and 3.58, while the decay ratio remains virtually constant around 0.21. As illustrated in [Figure B.1](#) (panels a and

b), normalizing the paths confirms that the geometric profile of the composite fee spike is invariant to the selection threshold. In contrast, asset returns vary monotonically with the strictness of the cutoff, validating [Fact 3](#). As the threshold tightens from none to $> 50\%$, the $t - 60$ to $t + 120$ raw return drops from -0.65% to -12.91% , and the return net of the S&P 500 declines from -8.40% to -19.57% .

Fee cutoff	Episodes	Fee			Return, $t-60$ to $t+120$	
		Peak (% APR)	Ramp (peak/ $t-60$)	Decay ($t+120$ /peak)	raw	net
<i>none</i>	1069	23.8	3.57	0.212	-0.65%	-8.40%
$> 1\%$	882	28.7	3.58	0.211	-1.54%	-9.63%
$> 5\%$	578	42.4	3.54	0.212	-3.07%	-11.24%
$> 10\%$	405	57.5	3.48	0.212	-7.22%	-14.86%
$> 20\%$	228	91.1	3.35	0.212	-8.56%	-15.82%
$> 50\%$	110	154.6	3.46	0.198	-12.91%	-19.57%

Table B.2: Sensitivity to the fee cutoff. Each row rebuilds the event sample at a different threshold; [Figure B.1](#) plots the underlying paths. Returns run from $t-60$ to $t+120$ (matching the decay resolution). Both raw and net S&P 500 returns fall monotonically as the cutoff tightens, validating [Fact 3](#). Source: `code/export_threshold_sensitivity.m`, with the fee columns reproduced by `gmod/robustness/unfiltered_event_study.py` and the price columns by `gmod/robustness/verify_cutoff_panels.py`.

B.4 Indicative Quotes and the Fee Moments

The data records indicative fees rather than actual contracted loan rates or volumes. However, because both fee moments are calculated as ratios of these quotes (ramp at $57.46/16.51$, decay at $12.18/57.46$), any systematic proportional bias across the listings cancels out, leaving these benchmarks unaffected even if the absolute fee levels are distorted. The return moments are measured from stock prices rather than from quotes, so a bias in the quotes reaches them only through which episodes clear the cutoff and which day is dated $t = 0$.

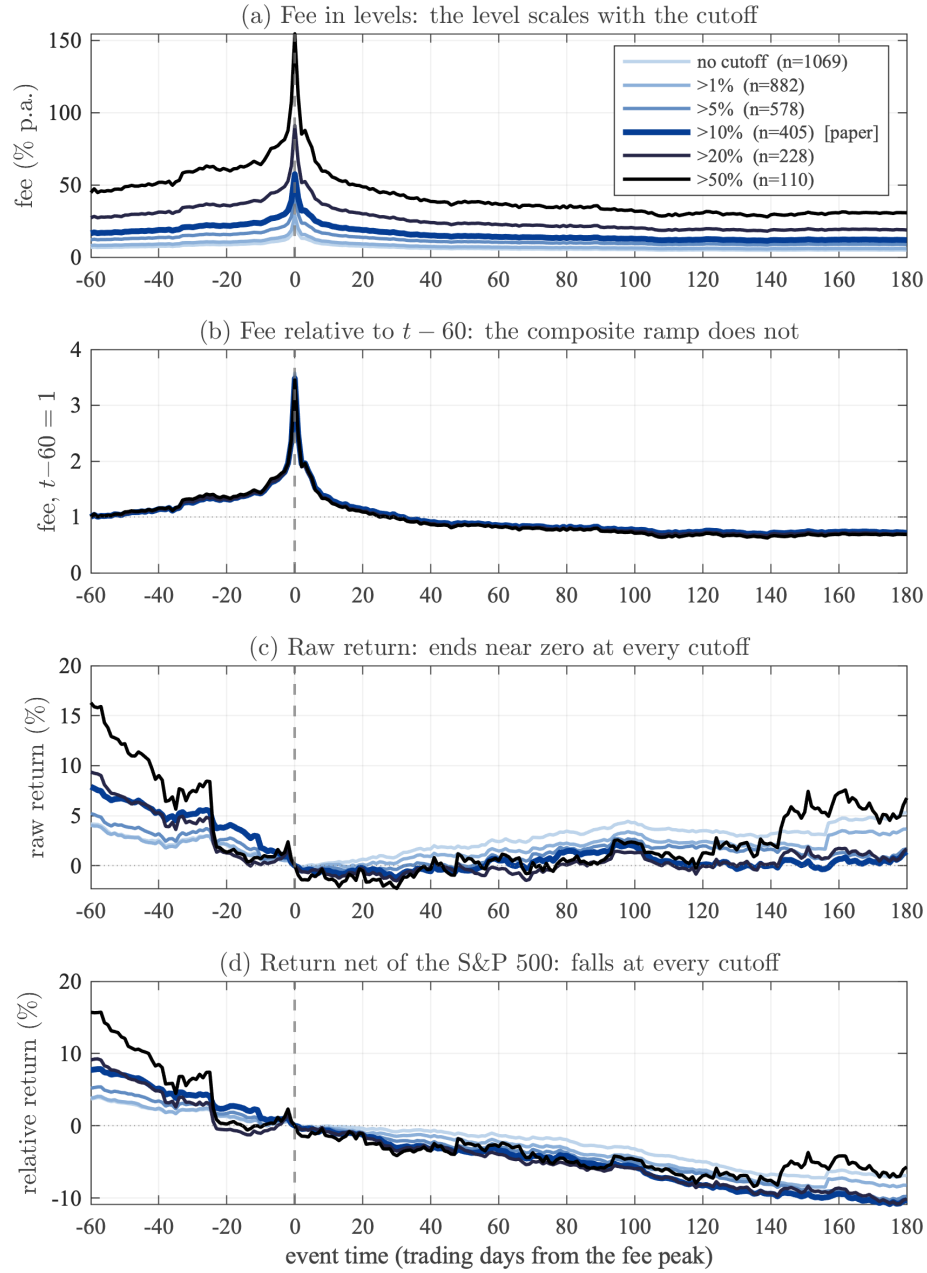


Figure B.1: Fee and return composites across cutoff thresholds. Each curve rebuilds Figure 1 on the episodes whose fee peak clears the stated threshold. Darker shading indicates tighter cutoffs; the heavy line is the 10% baseline. Panel (a) shows fee levels, panel (b) the same paths normalized to $t = -60$. Panels (c) and (d) plot the price against its fee-peak day level, in per cent, raw and net of the S&P 500. Source: `code/plot_fee_return_cutoffs.m`.

C Numerical Implementation

This appendix details the computational steps that solve the global equilibrium. The toolbox maps the structural conditions into a system of equations evaluated at every point of the discretized state space, stated below, together with the code that implements it.

C.1 Non-linear equilibrium system

The algorithm solves for a vector of eight unknowns,

$$\mathbf{X} = \{ p^s, c^P, c^O, a', p^\theta, \tilde{\mu}^{O,\text{cap}}, \tilde{\mu}^{O,s}, \tilde{\mu}^{O,\ell} \},$$

where p^s is the share price, c^i is agent i 's consumption, a' is the forward state choice, p^θ is the base rate, and the remaining three are the shadow prices of the Optimist's boundary constraints. Given the choice $a^{P'}$, individual portfolios are recovered using the smooth projection operators $\theta' = \text{neg}_\varepsilon(a')$ and $s^{O'} = \text{pos}_\varepsilon(1 - a')$, which keep the Newton step differentiable at the kink.

The continuation values of [Definition 2](#) enter the system,

$$V^i = \mathbb{E}^i \left[\beta \left(\frac{c^{i'}}{c^i} \right)^{-\sigma} (p^{s'} + d') \right], \quad i \in \{P, O\},$$

a prime denoting the next-period value and the expectation being taken under agent i 's own subjective belief matrix Π^i .

The core system consists of eight equations, solved at every point of the state grid to a

residual below 10^{-8} (`TolSol` = `1e-8`, the toolbox default):

$$\begin{aligned}
\text{eq1} \quad & -p^s - \gamma \theta' \text{neg}'_\varepsilon(a') - p^\theta \text{neg}'_\varepsilon(a') + V^P = 0 \\
\text{eq2} \quad & -(p^s - \kappa \tilde{\mu}^{O,\text{cap}}) + V^O + \tilde{\mu}^{O,s} = 0 \\
\text{eq3} \quad & p^\theta + \tilde{\mu}^{O,\ell} - \tilde{\mu}^{O,\text{cap}} = 0 \\
\text{eq4} \quad & (p^s + d)a + Y^P - p^s a' - p^\theta \theta' - \frac{1}{2}\gamma(\theta')^2 - c^P = 0 \\
\text{eq5} \quad & (p^s + d)(1 - a) + Y^O - p^s(1 - a') + p^\theta \theta' + \chi \frac{1}{2}\gamma(\theta')^2 - c^O = 0 \\
\text{eq6} \quad & \tilde{\mu}^{O,s} s^{O'} = 0 \\
\text{eq7} \quad & \tilde{\mu}^{O,\ell} \theta' = 0 \\
\text{eq8} \quad & \tilde{\mu}^{O,\text{cap}} - \alpha \text{neg}_\varepsilon(\kappa s^{O'} - \theta') = 0
\end{aligned}$$

The eighth is the penalty-based lending constraint approximation of [Section 5.4](#).

C.2 Code: GDSGE

Three tolerances, and bounds.

Quantity	Code	Value	What it bounds
Policy revision	<code>TolEq</code>	10^{-6}	largest change in c^P , c^O , p^s between sweeps
Equation residual	<code>TolSol</code>	10^{-8}	largest residual at any grid point
Share clearing	n/a	10^{-6}	smoothing error, grid-wide (Appendix D.5)

The first is the convergence criterion for the outer policy iteration and is the only one the solver stops on. The second is enforced point by point: any grid point whose residual exceeds it is re-solved. The third is not a setting at all but an accuracy check reported after the fact, and it measures the accounting cost of replacing the projections by their smoothed versions.

Notation in the listing. Each of these is recovered from the solved policy a' :

Object	Code	Recovered as
Pessimist's gross short $\theta^{P'}$	thetaPn	$\text{neg}(a')$
Pessimist's gross long $s^{P'}$	—	$\text{pos}(a')$
Optimist's net position $a^{O'}$	a0n	$1 - a'$
Optimist's gross long $s^{O'}$	s0n	$\text{pos}(1 - a')$
Optimist's lending $\ell^{O'}$	—	equals $\theta^{P'}$ by clearing
Slope of the short projection	neg_prime	$\text{neg}'_{\varepsilon}(a')$
Cap slack x	cap_slack	$\kappa s^{O'} - \theta^{P'}$
Cap penalty	cap_pen	neg_{ε} of the cap slack
Continuation values V^i	beta*E.i	E_i is V^i/β , $i \in \{P, O\}$

Source. `shorting_fee_cost.gmod`.

```

1  % Toolbox options
2  SIMU_RESOLVE = 0; SIMU_INTERP = 1;
3  PrintFreq = 10;
4  SaveFreq   = inf;
5  TolEq      = 1e-6;
6  MaxIter    = 500;
7
8  % Parameters
9  parameters beta sigma Pi_P Pi_0 psl psh kappa gamma_short eps_smooth alpha_cap chi;
10
11 beta        = 0.95;
12 sigma       = 1.5;
13 kappa       = 0.5;
14 gamma_short = 0.30;
15 eps_smooth  = 1e-4;
16 alpha_cap   = 10;
17 chi         = 0.40;
18

```

```

19 % 5-state discretized AR(1) dividend, rho = 0.9, sigma_eps = 0.01
20 var_shock d Y_P Y_0;
21 shock_num = 5;
22 d = [0.117646 0.146323 0.175000 0.203677 0.232354];
23 Y_P = 0.5*ones(1,5);
24 Y_0 = 0.5*ones(1,5);
25
26 ps1 = 1e-10*min(d)*beta/(1-beta);
27 psh = max(d)*beta/(1-beta);
28
29 % State: aP in [-1, 1], the feasibility wall -kappa/(1-kappa) at kappa = 0.5
30 var_state aP;
31 aPmin = -1;
32 aPmax = 1;
33 aP = linspace(aPmin, aPmax, 200);
34
35 Pi_P = [ ...
36     0.814466 0.185440 0.000092 0.000001 0.000001;
37     0.088248 0.792061 0.119662 0.000028 0.000001;
38     0.047627 0.072190 0.807991 0.072190 0.000001;
39     0.049636 0.000030 0.124729 0.825605 0.000001;
40     0.050000 0.000001 0.000097 0.194713 0.755190 ];
41
42 Pi_0 = [ ...
43     0.755190 0.194713 0.000097 0.000001 0.050000;
44     0.000001 0.825605 0.124729 0.000030 0.049636;
45     0.000001 0.072190 0.807991 0.072190 0.047627;
46     0.000001 0.000028 0.119662 0.792061 0.088248;
47     0.000001 0.000001 0.000092 0.185440 0.814466 ];
48 shock_trans = Pi_P;
49
50 var_tensor cP_init c0_init;
51 cP_init = Y_P + d.*aP;

```

```

52 c0_init = Y_0 + d.*(1-aP);
53
54 % 8 unknowns: five prices and quantities, plus the Optimist's three multipliers
55 var_policy ps c0 cP aPn pTheta mu0s mu0l mu0c;
56
57 inbound ps      psl  2*psh;
58 inbound c0      1e-8 1.2;
59 inbound cP      1e-8 1.2;
60 inbound aPn     -3   3;
61 inbound pTheta  0    1.0;
62 inbound mu0s    0    1.0;
63 inbound mu0l    0    1.0;
64 inbound mu0c    0    1.0;
65
66 var_output ps cP c0 aPn pTheta mu0s mu0l mu0c;
67
68 var_interp c0_next cP_next ps_next;
69 initial cP_next cP_init;
70 initial c0_next c0_init;
71 initial ps_next 0;
72
73 cP_next = cP;
74 c0_next = c0;
75 ps_next = ps;
76
77 % Model: 8 equations
78 model;
79 % Smoothed corner labelling
80 thetaPn = 0.5*(sqrt(aPn*aPn + eps_smooth*eps_smooth) - aPn);
81 a0n      = 1 - aPn;
82 s0n      = 0.5*(sqrt(a0n*a0n + eps_smooth*eps_smooth) + a0n);
83 neg_prime = 0.5*(aPn/sqrt(aPn*aPn + eps_smooth*eps_smooth) - 1);
84

```

```

85  psn'  = ps_next'(aPn);
86  psdn' = psn' + d';
87  cPn'  = cP_next'(aPn);
88  cOn'  = cO_next'(aPn);
89
90  E_P = GDSGE_EXPECT{psdn' * ((cPn')^(-sigma)) | Pi_P} / ((cP^(-sigma)));
91  E_0 = GDSGE_EXPECT{psdn' * ((cOn')^(-sigma)) | Pi_0} / ((cO^(-sigma)));
92
93  % Pessimist Euler: carrying-cost wedge AND fee wedge, both via  $-(d \text{ bill}/d aPn)$ 
94  eq1 = - ps - gamma_short*thetaPn*neg_prime - pTheta*neg_prime + beta*E_P;
95
96  % the first-order condition in  $s^0$ , divided through by  $\lambda^0$ 
97  eq2 = -(ps - kappa*mu0c) + beta*E_0 + mu0s;
98
99  eq3 = pTheta + mu0l - mu0c;
100
101  % Budgets: the fee is a transfer in full, the carrying cost at the rate  $\chi$ 
102  eq4 = (ps+d)*aP      + Y_P - ps*aPn      - pTheta*thetaPn -
        0.5*gamma_short*thetaPn*thetaPn - cP;
103  eq5 = (ps+d)*(1-aP) + Y_0 - ps*(1-aPn) + pTheta*thetaPn +
        chi*0.5*gamma_short*thetaPn*thetaPn - cO;
104
105  %  $s^0 \geq 0$ , product against the GROSS long  $sOn$ , not the net position  $aOn$ 
106  eq6 = mu0s*sOn;
107
108  %  $l^0 \geq 0$ , product against the gross short, which clearing makes lending
109  eq7 = mu0l*thetaPn;
110
111  % Cap complementarity, PENALTY: the slack's sign is imposed nowhere else,
112  % so a product here would stop enforcing the cap altogether
113  cap_slack = kappa*sOn - thetaPn;
114  cap_pen   = 0.5*(sqrt(cap_slack*cap_slack + eps_smooth*eps_smooth) - cap_slack);
115  eq8 = mu0c - alpha_cap*cap_pen;

```

```
116
117 equations;
118     eq1;
119     eq2;
120     eq3;
121     eq4;
122     eq5;
123     eq6;
124     eq7;
125     eq8;
126 end;
127 end;
128
129 simulate;
130     num_samples = 100;
131     num_periods = 10000;
132     initial aP 0;
133     initial shock 3;
134     var_simu ps pTheta;
135     aP' = aPn;
136 end;
```

D Robustness Check

D.1 Existence of Carrying-cost Coefficient

γ	Iterations	Metric	Drift	$\theta_{\max}^P$	$p_{\max}^{\theta}$	Verdict
0.05	500	4.2×10^{-6}	-0.0021	1.0021	1.1×10^{-2}	no convergence
0.10	240	9.7×10^{-7}	+0.0006	0.9994	8.1×10^{-5}	equilibrium
0.20	241	9.7×10^{-7}	+0.0016	0.9957	3.1×10^{-5}	equilibrium
0.30	242	9.6×10^{-7}	+0.0016	0.9953	3.1×10^{-5}	equilibrium
0.40	242	9.8×10^{-7}	+0.0016	0.9969	3.1×10^{-5}	equilibrium
0.50	242	9.9×10^{-7}	+0.0001	0.9999	3.7×10^{-4}	equilibrium
0.60	500	7.8×10^0	-1.6220	2.2488	9.5×10^{-1}	diverges

Table D.1: Existence Across the Carrying-cost Coefficient. Everything else is held at [Table 2](#). Drift is how far inside the wall the policy’s lowest image lies, so a negative entry is a violation of clause (v) of [Definition 2](#). Source: `gmod/fee_cost/run_gamma_sweep.sh`.

D.2 Fee Ramp Saturation and Carrying Costs

γ	Ramp, by length of the run					Trough at $K = 200$
	$K = 22$	$K = 50$	$K = 100$	$K = 200$	$K = 400$	
0.10	1.63	2.08	2.64	3.25	3.69	-0.4493
0.20	1.82	2.26	2.63	2.81	2.84	-0.2047
0.30	1.96	2.38	2.63	2.70	2.70	-0.1329
0.40	2.07	2.46	2.64	2.67	2.67	-0.0991
0.50	2.17	2.53	2.66	2.68	2.68	-0.0795

Table D.2: Fee ramp saturation and carrying costs. Peak size premium over its pre-run level, for a K -period run of the highest dividend state in each re-solved economy of [Table 5](#), each started from its own stationary mean. Bold marks the calibrated row and entries above the event study’s 3.48. The last column is the deepest position reached at $K = 200$, against a wall of -1 . Source: `gmod/fee_cost/ramp_by_gamma.m`, which writes the numbers to `data/model/ramp_by_gamma.csv`.

D.3 The Scored Moments Across the Carrying Cost

γ	Episodes	Ramp (peak/ $t-60$)	Decay ($t+120$ /peak)	Return, raw ($t-60$ to $t+120$)
0.10	99	1.520	0.490	−6.3%
0.20	98	1.693	0.464	−5.0%
0.30	98	1.911	0.457	− 4.7%
0.40	98	2.000	0.435	−4.3%
0.50	99	2.058	0.437	−2.8%
Elasticity in γ	—	+0.19	−0.07	−0.51
Event study	405	3.481	0.212	−7.2%

Table D.3: The scored moments across the carrying cost. The model re-solved at each γ and its event study rebuilt from the simulated episodes. Bold is the calibrated economy, and the range is the interval over which an equilibrium exists (Appendix D.1). The three moments do not move together: a steeper charge lifts the ramp toward the event study and carries the return away from it, so no coefficient in the interval improves both. The elasticity is the log ratio of the endpoints over the log ratio of the coefficients. Source: `gmod/fee_cost/export_endogenous_by_gamma.m`, which writes the numbers to `data/model/endog_by_gamma.csv`.

D.4 Frictions vs. Equilibrium

Case	γ	Fee	Iterations	Metric	Drift	$\theta_{\max}^P$	Verdict
Neither friction	0.00	no	237	9.5×10^{-7}	−0.0227	1.0227	leaks
Fee only	0.00	yes	500	2.2×10^{-2}	−0.0030	1.0030	no convergence, leaks
Cost only	0.30	no	240	1.0×10^{-6}	+0.0020	0.9980	equilibrium
Fee and cost	0.30	yes	240	1.0×10^{-6}	+0.0020	0.9980	equilibrium

Table D.4: Equilibrium Existence Under Frictions. *Drift* is how far inside the feasibility wall of Lemma 2, $a' \geq -\kappa/(1-\kappa)$, the policy’s lowest value lies, so a negative entry violates clause (v) of Definition 2. *Leaks* in the verdict column means exactly that: the policy carries the state off the bottom of its own grid, to a short larger than the Optimist’s book can lend, so no compact set is invariant and the solved object is not an equilibrium. The carrying cost is necessary and sufficient for an equilibrium. The basic fee-only row did not converge. Source: `gmod/fee/run_frictions_2x2.sh`.

D.5 Smoothing Residuals

Section 5.2 replaces non-differentiable projections (`pos`, `neg`) with smoothed versions (`posε` and `negε` at $\varepsilon = 10^{-4}$) to ensure the inner Newton step is differentiable at corners, but introducing a minor error floor for accounting identities. This appendix confirms that the resulting share-clearing residual distortion, $|s^{P'} - \theta^{P'} + s^{O'} - 1|$, is negligible, measuring $< 10^{-6}$ grid-wide, $< 10^{-8}$ ergodically. Therefore, positions reconstruction via `posε` and `negε` causes immaterial discrepancy.

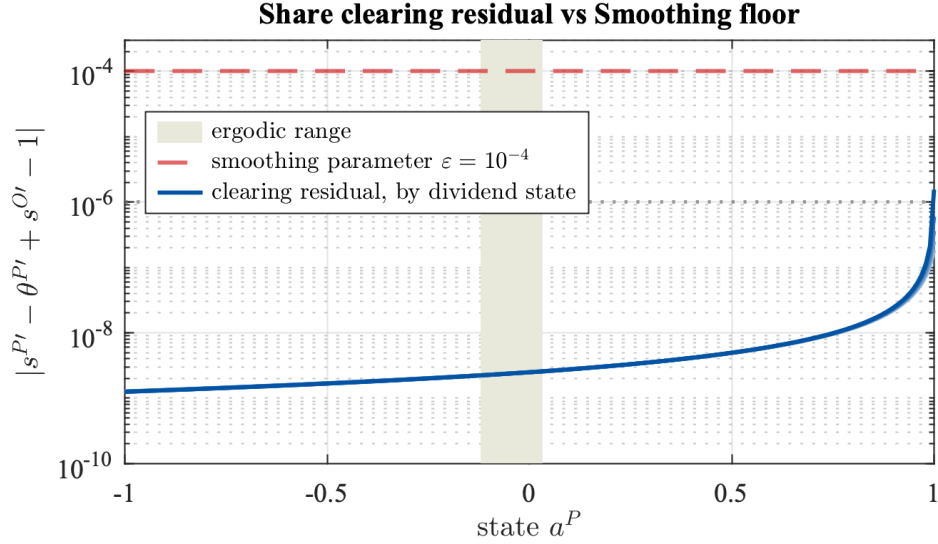


Figure D.1: Share Clearing Residual versus Smoothing Floor. The residual on a logarithmic axis, one line per dividend state, with the ergodic range shaded and the smoothing parameter $\varepsilon = 10^{-4}$ marked. It stays below 10^{-6} everywhere on the grid and below 10^{-8} across the whole ergodic range. Source: `gmod/fee_cost/plot_clearing_residual.m`.

E The Rest Point and the Stationary Mean

Two distinct summary statistics of the state dynamics appear in our results: the zero-drift rest point and the stationary mean. Both measures are constructed using the policy function $a'(z, a)$ and the *objective* transition matrix Π^P under which the economy is simulated.

Let $w(z)$ denote the stationary distribution of the objective state chain Π^P (the long-run share of time spent in each state). The left eigenvector $w = w\Pi^P$ is normalized so that $\sum_z w(z) = 1$. Its values are the weights reported in [Table 3](#). Averaging the state transition rule over this distribution defines the expected next-period state function $G(a)$ and the expected one-period drift $d(a)$:

$$G(a) = \sum_z w(z) a'(z, a), \quad \text{and} \quad d(a) = G(a) - a.$$

The *zero-drift rest point* a^* is defined as the root of this expected drift:

$$d(a^*) = 0 \quad \Longleftrightarrow \quad G(a^*) = a^*,$$

which is located via linear interpolation at the sign change of $d(a)$ along the state grid.

In contrast, the *stationary mean* $\bar{a}$ characterizes the ergodic distribution actually visited by the simulated economy. Let $D(z, a)$ be the joint stationary distribution of the state space,¹⁵ satisfying the forward invariant operator:

$$D(z', \hat{a}) = \sum_z \Pi^P(z, z') \sum_a D(z, a) \mathbf{1}\{a'(z, a) = \hat{a}\},$$

¹⁵On the solved grid $D(z, a)$ is a 5×200 array: five dividend states by two hundred position nodes. The nodes are evenly spaced, so $\Delta = 2/199$. The indicator below treats $a'(z, a)$ as landing on a node, which it almost never does. The weight λ that replaces it is mean-preserving, $\sum_a \lambda \hat{a} = a'(z, a)$ exactly. This matters here because $\bar{a}$ is itself a mean, and rounding to the nearest node would instead bias it by up to half a grid step at every iteration.

where, on the solved grid, the indicator is realised as the two-point weight

$$\lambda(z, a \rightarrow \hat{a}) = \max\left(0, 1 - \frac{|\hat{a} - a'(z, a)|}{\Delta}\right),$$

with Δ the grid spacing, which splits mass linearly between the two nodes bracketing $a'(z, a)$. Defining the marginal distribution of the agent's net asset position as $\mu(a) = \sum_z D(z, a)$, the stationary mean is given by $\bar{a} = \sum_a a \mu(a) / \sum_a \mu(a)$.

Thus, the rest point is the state where the drift *is* zero, while the stationary mean is the state where the drift *averages* to zero.

In the calibrated economy, the rest point is $a^* = -0.0408$ compared to a stationary mean of $\bar{a} = -0.0424$. The mean lies roughly 4% below the rest point because consecutive dividend shocks push the position short, from which it recovers only slowly due to endogenous frictions.

The choice between these two metrics depends on the specific economic question being addressed:

- **Local Stability:** [Section 6.5](#) and the drift/rest-point columns of [Table 5](#) employ a^* , given that the directional drift of a frozen position depends strictly on its location relative to a^* .
- **Average Aggregates:** [Section 7.3](#) and the corresponding lending premium columns employ $\bar{a}$, since the borrowing fee observed in a typical transaction reflects the pricing where the economy unconditionally spends its time.

F Frictionless: Non-Existence of Equilibrium

Set both frictions to zero, $\gamma = 0$ and $p_t^\theta \equiv 0$, and keep everything else: the same two agents, the same tree, the same beliefs, and the same inventory cap. The result is the frictionless benchmark. The iteration converges, but the policy it converges to carries the state past the feasibility wall of Lemma 2, violating clause (v) of Definition 2, so there is no recursive equilibrium.

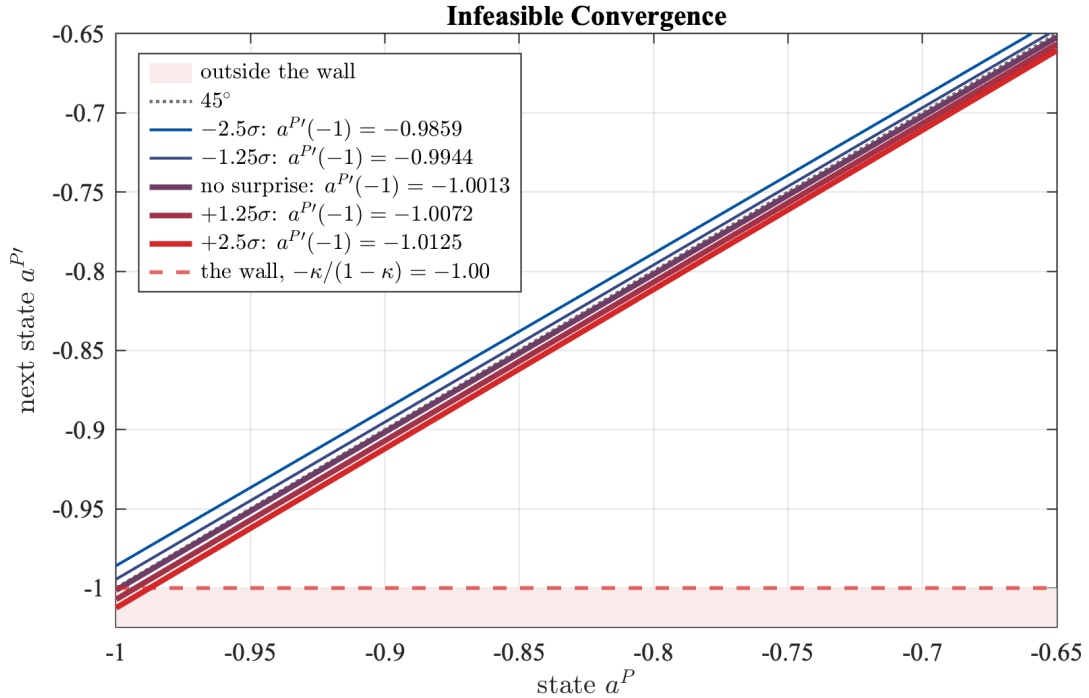


Figure F.1: Convergence Without Existence. The solved transition rule with both frictions off, one line per dividend state, near the left edge of the state space. The iteration converges, but three of the five states send the position below the feasibility wall $-\kappa/(1-\kappa) = -1$, shaded. A fixed point of the iteration need not be an equilibrium of the model. Source: `gmod/fee/plot_frictionless_leak.m`.